

Deception Aversion^{*}

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First version: 2nd June 2025

Current version: 22nd June 2026

Abstract

In communicating private information, opportunities to *lie* by misreporting the truth also present opportunities to *deceive* by inducing inaccurate beliefs. While many studies document truth-telling despite material costs—commonly attributed to lying aversion—such behavior may also reflect aversion to deceiving others. Disentangling the two preferences is challenging because deception depends on the sender’s unobserved second-order beliefs. In a novel game, we show theoretically how to identify deception aversion from choice data alone, under minimal assumptions on beliefs. In a laboratory experiment, we find strong evidence of deception aversion: many subjects lie to avoid deception; structural estimates imply that 30% are deception-averse.

Keywords: lying; deception; lying aversion; deception aversion; image concerns; strategic communication; psychological game theory

JEL Classification: C44, C72, C92

^{*}We thank Kai Barron, Juan Francisco Blazquiz-Pulido, Valeria Burdea, Syngjoo Choi, Tilman Fries, Alessandra Casella, Alessandro Ispano, Nicolas Jacquemet, Philippe Jehiel, Agne Kajackaite, Navin Kartik, Kiryl Khalmetski, Ernest Lai, Wooyoung Lim, Georgia Michailidou, Suanna Oh, Luca Polonio, Collin Raymond, Claire Rimbaud, Michael Thaler, and seminar participants at PSE, Indiana University, Purdue University, NYU-Abu Dhabi, NUS, UBC, SNU, Yonsei, ESWC-2025, SAET-2025, D-TEA-2026, the Barcelona Summer Forum PBR Workshop-2025, the Workshop on Narratives, Memory, and Beliefs-2025, and the Workshop on Bounded Rationality and Decision-making-2026 for helpful comments. This work was supported by the Agence Nationale de la Recherche (grant ANR-24-CE28-0028-01).

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Communicating private information is central to many economic and social activities: an expert gives advice (Crawford and Sobel [1982]), a prosecutor persuades a judge (Kamenica and Gentzkow [2011]), or a seller discloses information about product quality to a potential buyer (Grossman [1981]).

In standard economic theory, the messages available to the sender have no intrinsic meaning, and she will send whatever message maximizes her expected payoff. In contrast, a large experimental literature documents a pattern of *truth-telling* whereby subjects tend to tell the truth at the expense of their own material payoffs.

The leading explanation for truth-telling is simply that people are averse to *lying*—defined as the sending of messages that are not literally true. As our examples illustrate, however, what matters for the receiver is often not the literal meaning of messages, but the information conveyed by those messages. Hence, an alternative explanation is that people are averse to *deception*—defined as the sending of messages that induce inaccurate beliefs (Sobel [2020]). In this paper, we aim to disentangle the two preferences and, by so doing, provide novel evidence of *aversion to deceiving others*.

While lying and deception often coincide, the two may diverge whenever there is a separation between literal meaning and the information conveyed. This may arise through strategic interaction or, as is often the case with natural language, because the sender need not (or cannot) fully describe all payoff-relevant information. In such settings, a sender may tell the truth in order to deceive, or lie in order to avoid deception. Hence, lying aversion and deception aversion may generate very different patterns of communication, making the distinction crucial for predicting behavior and the design of institutions that rely on eliciting private information.¹

A fundamental challenge with studying deception aversion, or any belief-dependent preference, is that beliefs are unobservable. A sender deceives if she sends a message that *she believes* induces more inaccurate beliefs than another message she could have sent. Hence, the essential object is the sender’s *second-order beliefs*, the entire mapping

¹For example, in school choice, there is often a tradeoff between mechanisms that are *strategy-proof* (in which truth-telling is dominant) and those that may achieve higher *efficiency* in equilibria that involve strategic misreporting. If individuals are sufficiently lying-averse, they will instead report truthfully in the latter, which is likely to undermine efficiency gains. By contrast, if individuals are primarily deception-averse, this is not necessarily the case. Indeed, untruthful equilibrium messages may not be deceptive at all. Hence, the mechanism that works best in practice may be sensitive to the exact mix of lying and deception costs within the population.

from messages to her belief over the receiver’s belief (over the payoff-relevant state).

In this paper, we introduce a novel game and show theoretically how to identify deception aversion from choice data alone, with minimal assumptions on the sender’s second-order beliefs. In particular, we do not assume that beliefs are determined in equilibrium, nor do we elicit them directly. We run a laboratory experiment and find strong support for deception aversion: many subjects *lie in order to avoid deception*, and structural estimates imply that 30% of subjects are deception-averse.

The game we introduce is a variation of the standard paradigm from the lying aversion literature in which people privately observe the outcome of a *die roll* and are paid in proportion to the outcome they report (Fischbacher and Föllmi-Heusi [2013]).

Our version differs in two key ways. First, it is a sender-receiver (cheap-talk) game in which an outcome of the die is communicated by the sender to the receiver. Second, in addition to the die, the sender also privately observes the outcome of a *coin flip*, which determines the conditional distribution of die outcomes, with higher die outcomes being more likely following heads and less likely following tails. Hence, the message (which reports a die outcome only) may give information about the coin. After observing the message, the receiver reports a belief over the probability the coin is heads. She is incentivized for accuracy, and so the coin is payoff-relevant to the receiver.² The sender *lies* by misreporting the die and *deceives* by inducing inaccurate beliefs about the coin.

We consider two specifications for the sender’s payoffs, but in both cases, the sender communicates about the die, which influences beliefs about the coin. In the *message-payoff* game, the sender is paid in proportion to her own message (as in the standard paradigm). In the *belief-payoff* game, the sender is paid in proportion to the receiver’s reported belief (that the coin is heads). Hence, deception either comes as an externality of individually rational behavior, or is directly incentivized. These games are simple, and yet capture many common interactions, e.g., the seller of an object who manipulates the buyer’s beliefs over quality (the coin) by reporting signals of quality (the die).

We first ask: what are the behavioral signatures of deception aversion? To answer this, we consider a theoretical framework that allows for very general lying and deception costs. We assume only that (1) lying costs depend on both the die outcome and the sender’s message and that (2) deception costs depend on the coin outcome and the

²In the lab, the receiver is incentivized via the binarized scoring rule (Hossain and Okui [2013]).

second-order beliefs induced by the sender’s message. Importantly, we model second-order beliefs—the entire mapping from messages to the sender’s belief over the receiver’s belief about the coin—as an unobserved object upon which we impose no a priori restrictions.³ Within this framework, a sender who exhibits *coin effects*—an effect of the coin, conditional on the die—must be averse to deception.⁴

The intuition for why coin effects indicate deception aversion is straightforward. Conditioning on die outcome, any given message will induce the same belief in the receiver and result in the same material payoff and lying cost for the sender, independent of the coin. Hence, without deception costs, the sender’s message cannot be affected by the outcome of the coin. On the other hand, deception costs can lead to coin effects because the same message will be more or less deceptive, depending on the coin’s realization.

While our most general results do not impose any restriction on beliefs, it is natural for subjects to anchor beliefs on the naive assumption that senders have a tendency to tell the truth.⁵ Due to the positive correlation between the coin and die, such naïveté pushes first- and second-order beliefs to be monotonically increasing in the message. With increasing second-order beliefs, we show that deception-averse senders will exhibit *positive coin effects* whereby they send higher messages following heads as opposed to tails. Moreover, the size of the coin effect is predicted to increase in the degree of deception aversion. Hence, an individual’s *average coin effect* is a reduced-form statistic that allows us to summarize the heterogeneity of deception aversion in the population.

We find that many subjects exhibit positive coin effects (and very few have negative coin effects), consistent with deception aversion and increasing second-order beliefs.⁶ Within our framework, these subjects must be *lying in order to avoid deception*.

We formulate a simple structural model in which both lying and deception costs are linear. We jointly estimate the cost parameters by fitting the model to the data of individual sender-subjects. The model provides a very good fit for most subjects, capturing much of the heterogeneity in the data. Structural estimates imply that 18%

³Indeed, our only assumption on second-order beliefs is that they are measurable with respect to the receiver’s information set, which only contains the message observed.

⁴Similarly, a sender who exhibits *die effects*—an effect of the die, conditional on the coin—must be averse to lying.

⁵We formalize this simple idea with a level k -type model in the spirit of Wang et al. [2010].

⁶In the questionnaire that follows our main task, we elicit both first- and second-order beliefs and find that, indeed, both tend to be monotonically increasing in the message.

of subjects are averse to both lying and deception, 12% to deception only, and 37% to lying only. Hence, we find that a large fraction (30%) of subjects are deception-averse. Moreover, the estimated share of each type is similar across the *message-payoff* and *belief-payoff* games, suggesting robustness across a variety of strategic settings.

Finally, because deception does hurt the receiver, an important question is whether the effects we observe are due to an aversion to deception per se, or if they are fully driven by *social* or *other-regarding* preferences. To test this, we run an additional treatment based on the *pure lying* game in which the sender can lie in order to directly implement her desired sender-receiver allocation, but there is no opportunity to deceive. Using this game as a control, we argue that the deception aversion documented in other treatments cannot be fully driven by social preferences over final outcomes. We also find that coin effects are uncorrelated with measures of altruism, and that the distribution of coin effects is insensitive to reducing receiver-payoffs by a factor of 10.

In the remainder of this section, we review related literature. Section 1 introduces the theoretical framework. Section 2 gives the experimental design. Section 3 analyzes the beliefs data and introduces a family of models consistent with beliefs. Sections 4 and 5 give reduced-form results based on coin and die effects. Section 6 establishes that deception aversion cannot be fully driven by social preferences or other possible confounds. Section 7 presents the structural model and estimates, and Section 8 concludes.

Related literature. Our most novel findings concern deception aversion, on which there is little existing work. However, our game features opportunities to lie as well as to deceive, so we speak to the large and rapidly growing literature on lying aversion. Recent work by Gneezy et al. [2018] and Abeler et al. [2019] conclude that the typical findings from “die rolling” experiments can be explained by a model with (1) a fixed cost of lying and (2) image concerns—i.e., a cost from being *perceived as a liar*.⁷ Our theoretical framework is general enough to incorporate both types of lying costs, as well as others found in the literature.⁸ This allows us to conclude that our finding of “coin effects,” while explicable by deception aversion, cannot be explained by lying aversion.

⁷Khalmetski and Sliwka [2019] propose a cost of sending a message that is proportional to the equilibrium fraction of liars that send that message. Dufwenberg and Dufwenberg [2018] instead posit a cost that is proportional to the expected size of the lie, also determined in equilibrium.

⁸Our general notion of lying costs also allows for the third family of costs considered in the meta-study of Abeler et al. [2019]—costs based on social norms or comparisons (e.g. Gibson et al. [2013]).

Our definitions of lying and deception follow [Sobel \[2020\]](#), who only recently formalized these as distinct notions. Previous work has largely focused on studying the relationship between lying aversion and payoff consequences. [Gneezy \[2005\]](#) considers experimental sender-receiver games in which the sender has the opportunity to lie about which of two actions yields the receiver a higher payoff.⁹ He finds that the fraction of subjects willing to lie depends on payoff consequences, both their personal benefit and the harm caused to others.¹⁰ The earlier literature variously uses the terms “lying” and “deception,” but does not make a distinction between them. While [Gneezy \[2005\]](#) refers to “deception,” it is much closer to what [Sobel \[2020\]](#) (and we) refer to as lying. In a setup based on [Gneezy \[2005\]](#), [Sutter \[2009\]](#) pre-empts the modern distinction between lying and deception: by eliciting the sender’s beliefs over the receiver’s action following messages, he concludes that some senders tell the truth in order to deceive.¹¹

Only recently, in a purely theoretical paper, [Eilat and Neeman \[2023\]](#) were the first to formalize a notion of deception costs. Building upon [Sobel \[2020\]](#), who only defines deception with respect to *equilibrium* beliefs, they also take an equilibrium approach. In particular, they assume that only deviations from equilibrium messages are costly.¹²

In contrast, we define deception directly with respect to the sender’s second-order beliefs, making the natural assumption that deception costs depend on the state and the second-order belief induced about that state. By imposing only minimal assumptions on beliefs, our theory is agnostic about whether they are determined in equilibrium or they exist only in the mind of the sender. Hence, our results are compatible with equilibrium, but we do not require it for identification, which offers several empirical advantages.¹³

The paper most closely related to ours is [Choi et al. \[2025\]](#), which also experimentally disentangles lying and deception aversion.¹⁴ They do so in a two-stage reputation-

⁹[Gneezy et al. \[2013\]](#) study games in which “the decision to lie increases own payment independent of the counterpart’s decision, but potentially at a cost for the counterpart.”

¹⁰[Hurkens and Kartik \[2009\]](#) clarify possible interpretations of Gneezy’s findings.

¹¹[Blazquiz-Pulido et al. \[2024\]](#) combine a strategic sender-receiver game with a non-strategic version and more detailed belief elicitation to improve “the identification of the senders’ intentions to deceive.” [Alempaki et al. \[2025\]](#) show that some subjects deceive by being evasive rather than lying directly.

¹²Earlier work by [Kartik \[2009\]](#) characterizes equilibria in the presence of lying costs.

¹³The equilibrium approach does not allow for heterogeneous second-order beliefs and implies that beliefs are sensitive to the assumed distribution of deception costs. Hence, the *identification of* deception costs may depend on *assumptions over* deception costs.

¹⁴See also [Farina et al. \[2024\]](#), who argue that some subjects display patterns consistent with deception aversion in experimental verifiable disclosure games (in which it is possible to deceive, but not to lie).

building game in which senders, whose “preference type” is either aligned or misaligned with the receiver, send one of two messages about the payoff-relevant state.¹⁵ By making one type of sender a computer with a pre-programmed strategy, one variation of their game admits (non-deceptive) lying in equilibrium, and another admits (truthful) deception. First, they document deviations from equilibrium consistent with lying and deception aversion. However, classifying individual subjects requires identifying second-order beliefs through direct elicitation. We, on the other hand, introduce a novel game and an approach that does not require belief elicitation or an assumption of equilibrium.

By modeling a direct dependence of utility on second-order beliefs, we contribute to the field of psychological game theory (Geanakoplos et al. [1989]). In particular, deception aversion is related to guilt aversion (Battigali and Dufwenberg [2007] and Charness and Dufwenberg [2006]), an aversion to disappointing others’ payoff expectations, as opposed to inducing inaccurate beliefs. The experiments in this literature are typically based on trust, dictator, and ultimatum games and involve belief elicitation or directly giving information about the receiver’s expectations (e.g. Ellingsen et al. [2010]). More closely related to our work, Battigalli et al. [2013] show that the results of Gneezy [2005], previously explained in terms of lying aversion, are also consistent with guilt aversion under some mild assumptions on the sender’s second-order beliefs. In contrast, our games and identification strategy allow us to disentangle lying and deception aversion.

1 Theory

1.1 A game with lying and deception opportunities

A sender has 2-dimensional private information, which can be thought of as the outcome of a *coin* Θ and a *die* D . We use $\theta \in \{0, 1\}$ for realizations of Θ and $d \in \{1, 2, \dots, T\}$ for realizations of D . We let $p_{\Theta, D}(\theta, d)$, $p_{\Theta}(\theta)$, and $p_D(d)$ denote the joint and marginal probability mass functions (PMFs), respectively, which we assume have full support. Letting $p_{D|\theta}(d)$ be the PMF of D conditional on $\Theta = \theta$, we assume that the likelihood ratio $\frac{p_{D|1}(d)}{p_{D|0}(d)}$ is strictly increasing in d . Hence, the coin and die are *positively correlated*.

After observing the coin and die, the sender sends a message $m \in \{1, 2, \dots, T\}$ to the

¹⁵Their setup recalls Ettinger and Jehiel [2010] and Ettinger and Jehiel [2021], who theoretically predict and experimentally validate the use of a “deceptive tactic” in similar games.

receiver, which we think of as declaring an outcome of the die (though this need not be truthful). Hence, her strategy $\sigma_S : \{0, 1\} \times \{1, 2, \dots, T\} \rightarrow \Delta\{1, 2, \dots, T\}$ gives her probability of sending each message as a function of her private information. We use $\sigma_S(m|\theta, d)$ for the probability message m is sent after realizing (θ, d) . After observing the message, the receiver reports a (first-order) belief $b \in [0, 1]$ over the probability $\Theta = 1$, so that her pure strategy is $\sigma_R : \{1, 2, \dots, T\} \rightarrow [0, 1]$.

The receiver’s payoffs are such that, if she believes $\Theta = 1$ with probability b' , it is strictly optimal to report $b = b'$ (and hence the restriction to pure strategies is immaterial). We abstract from the details here, but in our laboratory implementation, receiver’s payoffs are based on the binarized scoring rule (Hossain and Okui [2013]).

We consider two different versions of the game, defined by differences in the sender’s payoffs. In the *message-payoff* game, the sender’s payoff $u^M(m)$ depends *only* on the message m she sends and is strictly increasing in m . In the *belief-payoff* game, the sender’s payoff $u^B(b)$ is strictly increasing in the receiver’s reported belief b . In both cases, the coin is payoff-relevant for the receiver, but *not* the sender.

Economic applications. These games are two natural variations that capture many common economic interactions. In the belief-payoff game, the sender has a material incentive to convince the receiver that $\theta = 1$. Hence, there is a *direct incentive* to deceive. This captures the seller of an object who manipulates the buyer’s beliefs over quality (the coin) by reporting signals of quality (the die). The message-payoff game is a simplified variation in which the sender’s payoff increases in the message sent, and therefore does not depend on the receiver’s action (as in the standard “die rolling” paradigm). Unlike in the belief-payoff game, deception comes only as an *externality*.

Experimentally, the comparison of the two games is interesting, providing insight into the role of strategic forces in modulating deception aversion. Importantly, however, our general strategy for identifying lying and deception aversion, outlined in Section 1.3, is the same for both games (and, in fact, many other variations; see Remark 3).

1.2 The equilibrium benchmark

We first establish the Perfect Bayesian equilibrium (PBE) benchmark. In both message-payoff and belief-payoff games, all PBE are pooling: no information is transmitted and the receiver does not update her beliefs after any (on-path) message.

Proposition 1. (i) In all PBE of the message-payoff game, the sender always sends the highest message and the receiver does not update beliefs (on path): $\sigma_S(T|\theta, d) = 1$ for all (θ, d) and $\sigma_R(T) = p_\Theta(1)$. (ii) In all PBE of the belief-payoff game, $(\theta = 0)$ -senders and $(\theta = 1)$ -senders use strategies that induce the same distribution over messages and the receiver does not update beliefs (on path): $\sum_d \sigma_S(m'|0, d)p_{D|0}(d) = \sum_d \sigma_S(m'|1, d)p_{D|1}(d)$ for all m' , and $\sigma_R(m') = p_\Theta(1)$ for any m' in the support of $\sigma_S(\cdot)$.

Proof. See Appendix 9.1. □

With lying and/or deception costs, there may be equilibria in which information is transmitted. One approach taken in the literature is to characterize equilibria in the presence of lying costs (e.g. Gneezy et al. [2018]) or, more recently, deception costs (Eilat and Neeman [2023]). In this paper, however, we favor a non-equilibrium framework that allows us to identify deception aversion on the individual subject level under minimal assumptions on the sender's second-order beliefs.

1.3 A non-equilibrium framework with lying and deception costs

We let $\beta : \{1, \dots, T\} \rightarrow [0, 1]$ be the sender's *second-order beliefs*, so that $\beta(m) \in [0, 1]$ is the sender's belief over the receiver's belief of the probability that $\Theta = 1$ after observing message m . For now, we assume that the sender believes that the receiver forms belief $\beta(m) \in [0, 1]$ with probability one, an assumption we later relax.¹⁶ We take β as exogenous and initially impose *no restriction* whatsoever on this function.¹⁷ There is of course the implicit assumption that β is measurable with respect to the receiver's information set, which only contains the message. Importantly, we as experimenters observe coin-die realizations (θ, d) , but *not* second-order beliefs β .

Following Sobel [2020], the sender *lies* if message m does not coincide with die outcome d . The sender *deceives* by sending message m if (she believes that) the message does not lead to the most accurate belief about the payoff-relevant state Θ . Unlike lying, deception depends on second-order beliefs.

¹⁶In Remark 6, we generalize second-order beliefs so that the sender may believe that the receiver's beliefs are drawn from a distribution: $\beta : \{1, \dots, T\} \rightarrow \Delta[0, 1]$. All results go through as stated, except with respect to a notion of *expected* deception costs, so the current restriction is purely for simplicity.

¹⁷We do not even impose that second-order beliefs are consistent with Bayes rule for *any* sender strategy. For example, we allow for $\beta(m') = 1$ for all m' , which is not consistent with any strategy.

Definition 1. Suppose the sender observes (θ, d) . The sender *lies* at (θ, d) if $m \neq d$. The sender *deceives* at (θ, d) if $m \notin \underset{m' \in \{1, \dots, T\}}{\operatorname{argmin}} |\beta(m') - \theta|$.

Lying and deception are conceptually distinct. In particular, telling the truth may be deceptive, and lying need not be.

Remark 1. Telling the truth at (θ, d) is deceptive if $d \notin \underset{m' \in \{1, \dots, T\}}{\operatorname{argmin}} |\beta(m') - \theta|$. Lying at (θ, d) is not deceptive if $m \in \underset{m' \in \{1, \dots, T\}}{\operatorname{argmin}} |\beta(m') - \theta|$.

We now posit a utility function for the sender, incorporating psychological costs to lying and deception.

Lying costs depend on die realization d and message m through the function $c^{lie}(m, d)$. This is extremely general, allowing for a fixed cost of lying, costs that increase in the distance between d and m , and costs that cannot be captured by conventional notions of distance. This also allows for image concerns—i.e. a cost from being *perceived as a liar* (e.g. Gneezy et al. [2018] and Abeler et al. [2019]).¹⁸ Referring to all of these costs collectively as “lying costs,” our goal is not to distinguish between the different components of lying costs, but to separate all such costs from deception costs.

Deception costs depend on coin realization θ and the induced second-order belief $\beta(m)$ through the function $c^{dec}(\beta(m), \theta)$. This is very general, though for some results, we will specialize to the natural case that costs increase in the *size of deception* $|\beta(m) - \theta|$.

Letting $u^g(m; \beta)$ be the expected material payoff in game $g \in \{M, B\}$, i.e. $u^g(m; \beta) = u^M(m)$ in message-payoff or $u^g(m; \beta) = u^B(\beta(m))$ in belief-payoff,¹⁹ the sender’s overall *subjective* utility is the material payoff minus the two types of psychological costs:

$$\phi^g(m, \theta, d; \beta) = u^g(m; \beta) - c^{lie}(m, d) - c^{dec}(\beta(m), \theta). \quad (1)$$

We assume that senders maximize utility, and so we let $\sigma_S^*(\theta, d) := \underset{m' \in \{1, \dots, T\}}{\operatorname{argmax}} \phi^g(m', \theta, d; \beta) \subset \{1, \dots, T\}$ be the set of optimal (pure) messages following coin-

¹⁸Let $\Lambda : \{1, \dots, T\} \rightarrow [0, 1]$ be the sender’s exogenously specified *second-order perception* of being a liar such that $\Lambda(m)$ is the sender’s belief of the receiver’s belief that $m \neq d$ after observing message m . Given Λ , we may suppose that $c^{lie}(m, d) = g(m, \Lambda(m), d)$ for some function g . Similarly, we can also incorporate costs based on social norms or comparisons (e.g. Gibson et al. [2013])—i.e. feeling less bad about lying if others are also lying—by defining a subjective belief over others’ lying behavior.

¹⁹This assumes that the sender believes the receiver best responds by reporting her belief truthfully.

die realization (θ, d) (with dependence on β and g suppressed). For simplicity, we assume a unique best response so that σ_S^* is *single-valued*, which holds generically.²⁰

Our first and most basic result shows how to identify the existence of deception aversion from choice data alone, without any assumption on second-order beliefs. We will later refine the result to make much more precise predictions (that we take to the data), but this weaker form establishes the main conceptual insight.

Proposition 2. *If $\sigma_S^*(0, d) \neq \sigma_S^*(1, d)$ for some d , then the sender is strictly averse to deception: either $c^{dec}(\beta(\sigma_S^*(0, d)), 1) > c^{dec}(\beta(\sigma_S^*(0, d)), 0)$, $c^{dec}(\beta(\sigma_S^*(1, d)), 0) > c^{dec}(\beta(\sigma_S^*(1, d)), 1)$, or both.*

Proof. Suppose for purposes of contradiction that $c^{dec}(\beta(\sigma_S^*(0, d)), 1) \leq c^{dec}(\beta(\sigma_S^*(0, d)), 0)$ and $c^{dec}(\beta(\sigma_S^*(1, d)), 0) \leq c^{dec}(\beta(\sigma_S^*(1, d)), 1)$. By definition of $\phi^g(\cdot)$, we have that $c^{dec}(\beta(\sigma_S^*(0, d)), 1) \leq c^{dec}(\beta(\sigma_S^*(0, d)), 0) \implies \phi^g(\sigma_S^*(0, d), 1, d) \geq \phi^g(\sigma_S^*(0, d), 0, d)$. By definition of $\sigma_S^*(\cdot)$, we have that $\phi^g(\sigma_S^*(1, d), 1, d) > \phi^g(\sigma_S^*(0, d), 1, d)$ and thus that $\phi^g(\sigma_S^*(1, d), 1, d) > \phi^g(\sigma_S^*(0, d), 0, d)$. A symmetric argument yields $\phi^g(\sigma_S^*(0, d), 0, d) > \phi^g(\sigma_S^*(1, d), 1, d)$, a contradiction. $\square$

Hence, any *coin effect*—an effect of the coin on the message, conditional on die outcome—indicates deception aversion.²¹ The intuition is as follows. Conditioning on die outcome, any given message will induce the same belief in the receiver and result in the same material payoff and lying cost for the sender, independent of the coin. Hence, without deception costs, the sender’s message cannot be affected by the outcome of the coin. On the other hand, deception costs can lead to coin effects because the same message will be more or less deceptive, depending on the coin’s realization.

Coin effects illustrate the dichotomy between lying and deception. Within our framework, a sender who exhibits coin effects is deception-averse and yet tells lies.²² She therefore *lies in order to avoid deception*.

Remark 2. Proposition 2 shows that coin effects are sufficient for identifying deception aversion. They are not, however, necessary. For instance, a sender with perfectly “flat”

²⁰For any (θ, d) , for almost all (c^{lie}, c^{dec}) , (1) has a unique maximizer for almost all β .

²¹Technically, at this level of generality, coin effects may indicate that the sender is deception *loving*, so it is more accurate to say “non-neutral deception preferences.” For later results, we will refine the deception cost function to be able to identify deception aversion specifically.

²²If $\sigma_S^*(0, d) \neq \sigma_S^*(1, d)$ for some d , then necessarily either $\sigma_S^*(0, d) \neq d$, $\sigma_S^*(1, d) \neq d$, or both.

second-order beliefs ($\beta(m) = \beta(m')$ for all m, m') is incapable of deception, and so will not exhibit coin effects even if she is extremely deception-averse. Hence, our experiment identifies a *lower bound* on the fraction of subjects who are deception-averse.

In a similar fashion, we identify lying aversion. Any *die effect*—an effect of the die on the message, conditional on the coin outcome—indicates lying aversion.

Proposition 3. *If $\sigma_S^*(\theta, d) \neq \sigma_S^*(\theta, d')$ for some θ and $d \neq d'$, then the sender is strictly averse to lying: either $c^{lie}(\sigma_S^*(\theta, d), d') > c^{lie}(\sigma_S^*(\theta, d), d)$, $c^{lie}(\sigma_S^*(\theta, d'), d) > c^{lie}(\sigma_S^*(\theta, d'), d')$, or both.*

Proof. See Appendix 9.1. The proof closely parallels the proof of Proposition 2. $\square$

Hence, coin effects indicate deception aversion and die effects indicate lying aversion. A useful corollary states that purely deception-averse senders ignore the die, and purely lying-averse senders ignore the coin.

Corollary 1. *(i) A purely deception-averse sender (i.e. with no lying costs, $c^{lie}(m, d) = c^{lie}(m', d')$ for all m, m', d, d') will ignore the die: $\sigma_S^*(\theta, d) = \sigma_S^*(\theta, d')$ for all θ, d , and d' . (ii) A purely lying-averse sender (i.e. with no deception costs, $c^{dec}(\beta(m), \theta) = c^{dec}(\beta(m'), \theta')$ for all m, m', θ, θ') will ignore the coin: $\sigma_S^*(0, d) = \sigma_S^*(1, d)$ for all d .*

Senders who are averse to both deception and lying may exhibit both coin and die effects. Indeed, we find many such subjects in our experiment.

With additional structure on second-order beliefs and deception costs, we make more precise predictions. In particular, consider the following monotonicity assumption.

Assumption 1. *(1) Second-order beliefs $\beta(m)$ are strictly increasing in m , and (2) deception costs satisfy $c^{dec}(\beta(m), \theta) = f(|\beta(m) - \theta|)$ for (weakly) increasing f .*

Part (1) of the assumption states that second-order beliefs are strictly increasing. In Section 3, we show theoretically that this will be the case for naive senders who believe that receivers are sufficiently credulous. Moreover, at the end of our experiment, we elicit second-order beliefs and find that they do, in fact, tend to be increasing. Part (2) is a very weak assumption, stating that the cost of deception increases in the *size of deception*, as measured by the distance between θ and the induced belief $\beta(m)$. Because we only

assume that $f(\cdot)$ is weakly increasing, this allows for *no deception costs* whatsoever, a fixed cost of deceiving *at all*, or a fixed cost of deceiving above some threshold.

Under Assumption 1, the sender will tend to send higher messages when $\theta = 1$. That is, we should expect *positive coin effects*: $\sigma_S^*(1, d) - \sigma_S^*(0, d) \geq 0$. Moreover, if two senders have the same lying cost function and second-order beliefs, if one finds deception more costly, she will exhibit an even larger coin effect.

Proposition 4. *Suppose Assumption 1 holds. (i) $\sigma_S^*(1, d) - \sigma_S^*(0, d) \geq 0$ for all d . (ii) Let 1 and 2 be two senders with the same second-order beliefs β and lying cost function c^{lie} . If deception is more costly for 1 than 2,²³ then $\sigma_S^{*1}(1, d) \geq \sigma_S^{*2}(1, d)$, $\sigma_S^{*1}(0, d) \leq \sigma_S^{*2}(0, d)$, and $\sigma_S^{*1}(1, d) - \sigma_S^{*1}(0, d) \geq \sigma_S^{*2}(1, d) - \sigma_S^{*2}(0, d)$ for all d .*

Proof. See Appendix 9.1. □

To gain some intuition, consider the special case of a purely deception-averse sender (i.e. with no lying costs). Under Assumption 1, she will always send the highest message T when $\theta = 1$ because this both maximizes expected material payoffs and minimizes deception costs. On the other hand, when $\theta = 0$, there is a tradeoff: higher messages result in greater material payoffs but also incur greater deception costs. Thus, following $\theta = 0$, the costlier is deception, the lower is the message sent. In the extreme case, if deception is sufficiently costly, the sender will always message 1 following $\theta = 0$.

Corollary 2. *Suppose Assumption 1 holds. (i) A purely deception-averse sender (i.e. with no lying costs) will use the strategy $\sigma_S^*(1, d) = T$ and $\sigma_S^*(0, d) = x$ for all d and some x . (ii) Let 1 and 2 be two purely deception-averse senders with the same second-order beliefs β . If deception is more costly for 1 than 2, then $\sigma_S^{*1}(0, d) \leq \sigma_S^{*2}(0, d)$ for all d .*

To summarize, coin effects indicate deception aversion and die effects indicate lying aversion. With additional monotonicity assumptions on β and c^{dec} , any coin effects will have a positive sign whose magnitude increases in the degree of deception aversion.

The results of this section are quite general. Proposition 2, Proposition 3, and Corollary 1 depend only on separability—and even this can be relaxed to some extent.²⁴

²³Deception is more costly for 1 than 2 if $c_1^{dec}(\beta(m), \theta) = f(|\beta(m) - \theta|)$ and $c_2^{dec}(\beta(m), \theta) = g(|\beta(m) - \theta|)$ where, for every $\delta \geq 0$, $f(\delta + \epsilon) - f(\delta) \geq g(\delta + \epsilon) - g(\delta)$ for every $\epsilon \geq 0$.

²⁴For example, letting $\phi^g(m, \theta, d; \beta) = u^g(m, d; \beta) - c^{dec}(d, \beta(m), \theta)$ so that (1) material utility and lying costs are combined into one term and (2) deception costs are allowed to interact with the die outcome, coin effects would still indicate deception aversion.

Proposition 4 and Corollary 2 require the additional Assumption 1, but no results require any assumption on the shape of lying costs c^{lie} . We offer four remarks.

Remark 3. All results in this section go through under various generalizations of the game itself. First, the sender’s expected material utility $u(m; \beta)$ can be left completely general (i.e. any function of m , parameterized by β).²⁵ Second, because the sender’s optimal strategy does not depend on the joint distribution of Θ and D conditional on second-order beliefs, the results are also robust to arbitrary correlations between Θ and D . We are interested in the positive correlation case because, as we argue in Section 3, this gives rise to very regular second-order beliefs that typically satisfy Assumption 1.

Remark 4. An alternative formulation of deception costs would be $\tilde{c}^{dec}(\beta(m), \beta^*(\theta))$ where $\beta^*(\theta) := \beta(m'') : m'' \in \underset{m' \in \{1, \dots, T\}}{\operatorname{argmin}} |\beta(m') - \theta|$ is the *most accurate* belief the sender believes she can induce. However, our formulation $(c^{dec}(\beta(m), \theta))$ is behaviorally equivalent,²⁶ simplifies notation, and makes some results more transparent.

Remark 5. While we have only defined deception and deception costs with respect to the payoff-relevant state Θ , our notion of lying costs is general enough to also encompass deception costs with respect to the non-payoff-relevant state D .²⁷ Allowing for such costs, all results go through as stated, except we should replace “deception aversion” with “ Θ -deception aversion” (and “lying aversion” with “lying aversion and/or D -deception aversion”). In such case, our results allow us to separately identify Θ -deception aversion.

Remark 6. We have imposed that, after sending message m , the sender believes that the receiver will form belief $\beta(m) \in [0, 1]$ with probability one. This can be relaxed by allowing $\beta(m)$ to be a random variable supported on $[0, 1]$ with associated Borel measure $\mu(\cdot|m) \in \Delta[0, 1]$. Letting $\beta' \in [0, 1]$ be an arbitrary *realization* of $\beta(m)$, define *expected* deception costs $\bar{c}^{dec}(\beta(m), \theta) := \int c^{dec}(\beta', \theta) d\mu(\beta'|m)$. After generalizing Assumption 1,²⁸ all results go through as stated, except with $\bar{c}^{dec}(\beta(m), \theta)$ replacing $c^{dec}(\beta(m), \theta)$.

²⁵Corollary 2 requires, in addition, that $u(m; \beta)$ is strictly increasing in m .

²⁶Fix β , which induces $\beta^*(\theta)$. If $\beta(m) = \beta^*$ for all m , then $\beta^*(0) = \beta^*(1) = \beta^*$ and any $c^{dec}(\beta(m), \theta)$ and $\tilde{c}^{dec}(\beta(m), \beta^*(\theta))$ are behaviorally equivalent as they are constant in m for any given θ . If $\beta(m) \neq \beta(m')$ for some m, m' , simply set $c^{dec}(\beta(m), \theta) = \tilde{c}^{dec}(\beta(m), \beta^*(\theta))$ for all m and θ .

²⁷Let $\gamma : \{1, \dots, T\} \rightarrow \Delta\{1, \dots, T\}$ be the sender’s exogenously specified *second-order belief* over the receiver’s belief about D . Given γ , let $c^{lie}(m, d) = h(m, \gamma(m), d)$ for some function h .

²⁸**Assumption 1*.** (1) $m' > m \implies \beta(m') \succ_{f \circ sd} \beta(m)$, and (2) $\bar{c}^{dec}(\beta(m), \theta) = \int c^{dec}(\beta', \theta) d\mu(\beta'|m)$ where $c^{dec}(\beta', \theta) = f(|\beta' - \theta|)$ for (weakly) increasing f .

2 Experimental Design

The experiment consisted of three **deception treatments** (*message-payoff*, *message-payoff-\$1*, and *belief-payoff*) and one **control treatment** (*pure lying*). Each corresponds to a different version of a sender-receiver game, and each subject participated in exactly one treatment—a between-subject design. All sessions followed the same basic structure, with one section of 41 rounds, followed by a questionnaire involving belief elicitation.

Common structure of all treatments.

Private information. The sender’s private information is identical in all treatments. The sender privately observes (1) the outcome of a coin-flip—heads or tails—both equally likely and (2) the outcome of an 8-sided die roll. The coin and die are positively correlated, with higher die outcomes more likely after observing heads and less likely after observing tails. The conditional distributions of die outcomes following heads and tails are upper- and lower-triangular, respectively, as depicted in Figure 1. This information was communicated to subjects via this exact image as well as through tables. In terms of the theoretical framework of Section 1: $T = 8$; $\theta = 1$ corresponds to heads; $\theta = 0$ corresponds to tails; $p_{\Theta}(1) = \frac{1}{2}$; $p_{D|1}(d) = (8 + d)/100$; and $p_{D|0}(d) = (17 - d)/100$. By using an 8-sided die, as opposed to the more standard 6-sided, the triangular distributions can be expressed using whole number percentage-probabilities.

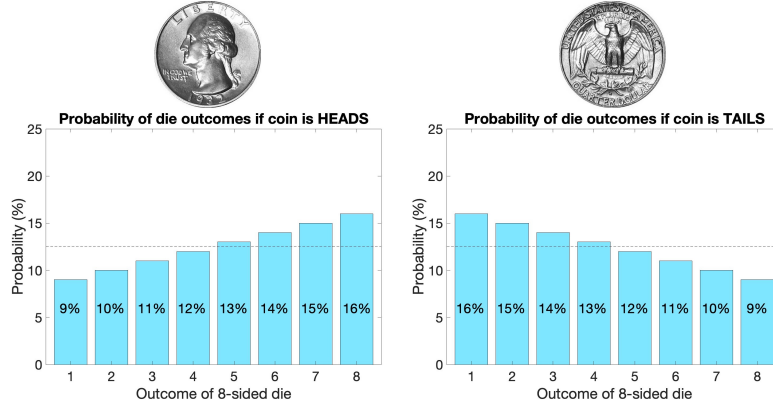


Figure 1: *Distribution of coin and die outcomes.* This figure depicts the conditional distributions (PMFs) of die outcomes following heads and tails, respectively.

Rounds 1-40. In each of the first 40 rounds, all subjects played in the role of the sender. In each round, subjects privately observed new coin and die outcomes (independent of previous rounds and of other subjects' outcomes). After observing this information, subjects sent a message stating "*The outcome of my die roll is _.*" where they were allowed to send any number 1 through 8. Importantly, messages were *not* immediately transmitted. It was only in round 41 that *one* of each subject's 40 messages was randomly selected and shown to another subject in the role of the receiver. Hence, there is no feedback and the games are strategically one-shot.

As the private information and action space of the sender are the same in all treatments, the experimental interface in rounds 1-40 was identical across treatments.

Round 41. In round 41, all subjects played in the role of the receiver. Each subject observed *one* message that was sent by a randomly selected (other) subject in a randomly selected previous round. Subjects were told "*A participant (other than you) and a round are randomly selected. In this round, the participant sent the message: 'The outcome of my die roll is _.'*" After observing this message, subjects were prompted to take a single action, which was to either (1) report a belief (incentivized) over the probability the coin was heads (in the deception treatments) or (2) simply click a button to acknowledge having observed the message (in the pure lying treatment).

Belief elicitation. Immediately after round 41, there were eight belief-elicitation questions, one for each of the eight possible messages. This was part of a section described to subjects as a "questionnaire," which also involved some additional questions.²⁹ The content of this section came as a surprise to subjects.

Some subjects were asked for their *first-order* beliefs, and some were asked for their *second-order* beliefs. The first-order belief questions were exactly as in round 41 of the deception treatments, except that beliefs about the coin were elicited following all eight messages. The second-order belief questions asked for beliefs over the average first-order beliefs of other subjects, again following all eight messages. Because first-order belief elicitation questions were exactly the same as in round 41 (and immediately followed it), they were not incentivized. On the other hand, as the concept of second-order beliefs had not been previously introduced to subjects, they were incentivized via the binarized scoring

²⁹These included free responses, questions about attitudes toward lying, questions to measure strategic sophistication and cognitive ability, and basic demographics. In some later sessions, we also included an incentivized dictator game, which was added in response to referee comments.

rule (Hossain and Okui [2013]).³⁰ All eight questions were presented on separate screens and in random order, with each screen resembling that in round 41 as closely as possible.

In the message-payoff treatment, we asked half the subjects for their *first-order* beliefs and the other half for their *second-order* beliefs. In all other treatments, we asked for first-order beliefs. Because beliefs are not reported in round 41 of the pure lying treatment, second-order beliefs are not well-defined there. Hence, this gives a common belief measure across treatments. In any case, elicited beliefs, whether first-order or second-order, are very similar both within and across all treatments (Section 3).

Message-payoff treatment (baseline). In the message-payoff game, the sender’s payoff depends only on the message that is randomly selected and shown to the receiver. The payment, which is in terms of a probability of earning \$10,³¹ is proportional to the message, as shown in Table 1. In terms of the theoretical framework, $u^M(m) = 0.32 + 0.04m$.

	Message							
	1	2	3	4	5	6	7	8
Sender’s payment:	36%	40%	44%	48%	52%	56%	60%	64%

Table 1: *Message-payoff game—sender’s payment.* This table gives the probability the sender earns \$10 as a function of the randomly selected message from rounds 1-40.

The receiver’s task and payment were as follows. After being shown a message in round 41, subjects were asked “*What is your guess for the probability this participant’s coin was HEADS in this round?*”. Subjects then reported their beliefs, which were incentivized via the binarized scoring rule, resulting in a probability of earning \$10.

Message-payoff-\$1 treatment. This treatment was exactly the same as the baseline message-payoff treatment, except with receiver-payoffs reduced by a factor of 10: the binarized scoring rule gives a chance at \$1 instead of \$10.

Belief-payoff treatment. In the belief-payoff game, the sender earns a higher payoff if the randomly selected message convinces the receiver that her coin was heads.

³⁰For each subject, one of the eight messages is randomly selected. The subject earns \$1 with a probability that depends on her reported second-order belief (corresponding to that message) and a randomly selected first-order belief of another subject from round 41 (who saw that message).

³¹All payments are in “probability points.” Note that, in order to use the binarized scoring rule, the receiver’s payoffs are necessarily in probability points, so this ensures that sender and receiver payoffs are in the same units.

Specifically, the sender earns \$10 with a probability that equals the receiver’s reported belief. In the language of the theoretical framework, $u^B(b) = b$.

We note that for senders who believe the receiver is Bayesian and credulous—i.e. believing every message is sent truthfully—her second-order beliefs become $\beta(m) = 0.32 + 0.04m$ and her subjective expected payoffs exactly coincide with those in the message-payoff game. We refer to this as the *truthful/Bayesian* benchmark.

The task and payment of the receiver are exactly as in the message-payoff game, based on reported beliefs and the binarized scoring rule. Hence, the screens shown to subjects in round 41 are identical across all deception treatments.

Pure lying treatment. The pure lying game is a *control game* that offers the opportunity to lie, but not to deceive: the sender has the same private information, but the coin is no longer payoff-relevant. Instead, the payoffs to both sender and receiver are a function of the randomly selected message. Hence, the sender directly determines her desired sender-receiver allocation, but she may have to lie in order to implement it.

	Message							
	1	2	3	4	5	6	7	8
Sender’s payment:	36%	40%	44%	48%	52%	56%	60%	64%
Receiver’s payment:	64%	60%	56%	52%	48%	44%	40%	36%

Table 2: *Pure lying game—sender and receiver payments.* This table gives the probabilities with which both sender and receiver earn \$10 as a function of the randomly selected message from rounds 1-40.

The exact payoffs are in Table 2. The sender’s payoffs are exactly as in the message-payoff game, such that a higher message yields a higher probability of earning \$10: $u^P(m) = 0.32 + 0.04m$. For the receiver, a higher message yields a *lower* probability of earning \$10, according to $u_R^P(m) = 0.68 - 0.04m$. Note that $u^P(m) + u_R^P(m) = 1$ for all m , but the realization of payments is independent across sender and receiver. Instead of reporting a belief, receivers are asked to simply click a button to acknowledge they observed the message. In this way, messages are transmitted to the other party in a similar way across treatments, an important consideration for making the treatments more comparable in terms of lying costs.

The pure lying game, in which deception is not possible, is used as a control to rule out that coin effects are driven by other factors, namely experimenter demand, entitlement effects, or social preferences (Section 6).

Logistical details. Our sessions were run at the Economic Research Laboratory (ERL) of Texas A&M University. Subjects were undergraduate students, recruited via the ORSEE software (Greiner [2015]). The message-payoff treatment—our baseline—had 144 subjects total. The other treatments, message-payoff-\$1, belief-payoff, and pure lying, had 74, 82, and 50 subjects, respectively. For half of the subjects in message-payoff (74), second-order beliefs were elicited; for all other subjects, first-order beliefs were elicited. Including a \$10 show-up fee, the average payment was \$22 for an experiment lasting about 50 minutes. Experimental instructions and screenshots are given in Online Appendices 10.6 and 10.7.

Discussion of the experimental design. A few features of our design merit discussion. Firstly, for the large majority of the experiment—the first 40 of 41 rounds—subjects play as the sender. We do this because we are primarily interested in sender strategies, due to our focus on documenting lying and deception aversion. With 40 rounds, we are able to have good estimates of individual-level coin effects, which require both heads and tails realizations for multiple die outcomes.³² Alternatively, we could have used the strategy method, but this may have led to experimenter demand or contrast effects, and would have been more difficult to explain to subjects.³³

It is important that exactly *one* message is transmitted to the receiver. If multiple messages are transmitted, then the sender might have incentive to create a “portfolio” of messages to appear more credible. Consider, for example, a subject who always messages 8—the payoff-maximizing choice in the message-payoff game. This would reveal to the receiver that the message is not very informative. Hence, if multiple messages are transmitted, it would likely lead to mixing and data that is hard to interpret: indeed, second-order beliefs would be a mapping from the entire set of transmitted messages.

Finally, our theoretical analysis depends on an “accepted meaning of messages,” in particular that messages are commonly understood as declaring a die outcome. It is for this reason that we force messages to take the form “*The outcome of my die roll is _.*” as opposed to simply reporting a number. In our view, this, and the fact that we do not provide feedback, makes adopting other languages implausible.

³²More than 90% of subjects observe at least 14 of 16 coin-die outcomes.

³³Our primary concern with the strategy method, or any method that may highlight that a message is to be sent following every coin-die outcome, is that it could encourage “contrast reasoning” that leads subjects to send different messages following different coin outcomes.

3 Beliefs

Before examining the empirical sender strategies, which is our focus, we analyze elicited beliefs. Establishing their general pattern will help us to interpret later results, but we emphasize that our approach does not rely on identifying each individual’s beliefs, with our main analysis based entirely on choice data.³⁴

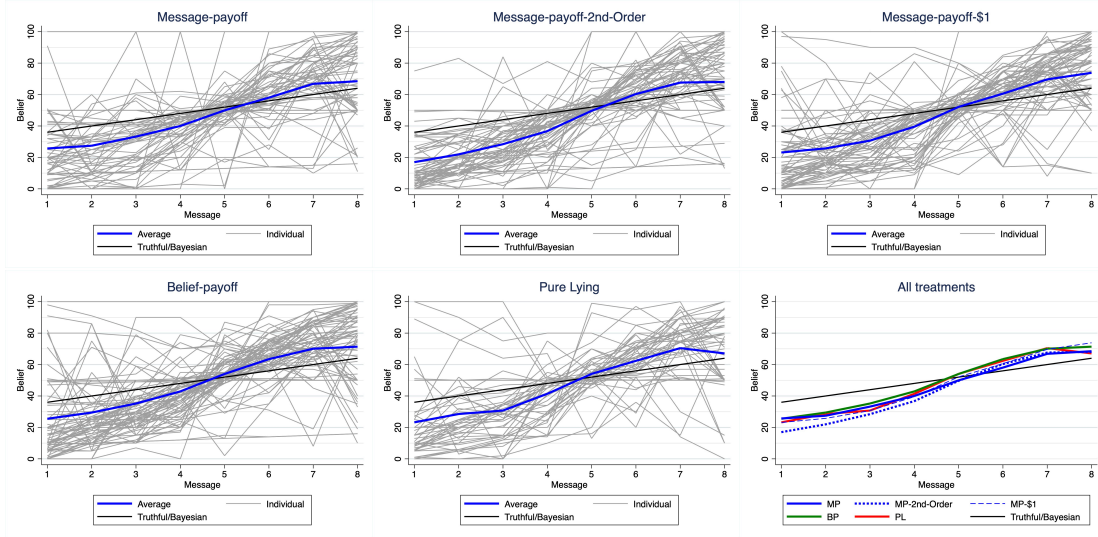


Figure 2: *Elicited beliefs.* The first five panels plot, for each treatment, all individual subjects’ elicited beliefs as reported in the eight questions of the questionnaire. Each panel gives the average belief (blue) and the truthful/Bayesian benchmark—the beliefs formed by a perfectly Bayesian subject who believes the sender always tells the truth. The bottom-right panel compares the average beliefs across all treatments.

Figure 2 plots elicited beliefs following each of the eight messages, based on the eight questions from the questionnaire. Each of the first 5 panels corresponds to a different treatment. Note that elicited beliefs correspond to (1) *second-order* beliefs in message-payoff-2nd-order and (2) *first-order* beliefs in the other treatments. The bottom-right panel shows that the average beliefs, whether first-order or second-order, are very similar across treatments. In Online Appendix 10.5, we show that they are statistically indistinguishable.

³⁴We favor an approach that does not rely on elicited beliefs because of the view that beliefs are fundamentally unobservable and elicited beliefs are simply measures of underlying beliefs. As such, elicited beliefs are subject to measurement error (i.e. noise) and possibly bias. Schotter and Trevino [2014] and Schlag et al. [2015] provide excellent discussions along these lines.

While there is substantial heterogeneity across subjects (as well as noise), a majority of subjects report beliefs that are perfectly or approximately monotonically increasing in the message.³⁵ In each treatment, the average beliefs are either perfectly monotonic or feature a slight dip in going from message 7 to message 8.

Inspecting individual subjects' beliefs more carefully, we find that while most subjects have approximately monotonic beliefs, about 17% of subjects have beliefs that are approximately *hump-shaped*. For most of these subjects, beliefs peak at 7, but for some the peak is at 6. We plot these subjects' beliefs in Figure 3 and find that their average beliefs peak at 7 in all treatments, with a very pronounced dip in going from 7 to 8. Indeed, for these subjects, the average belief following 8 is lower than that following 6 in all treatments.

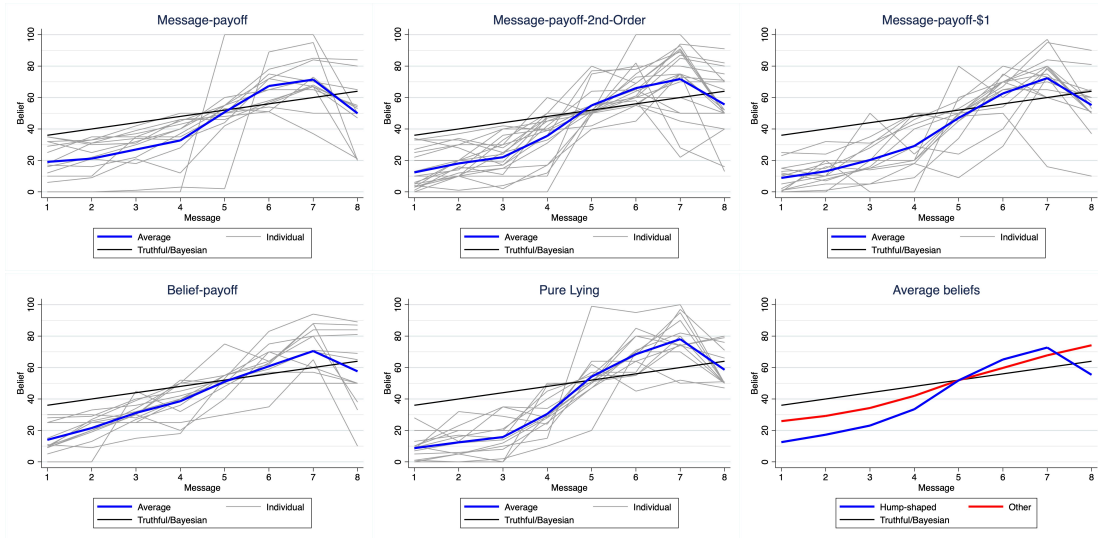


Figure 3: *Elicited beliefs of subjects with hump-shaped beliefs.* We plot the beliefs for the approximately 17% of subjects who have hump-shaped beliefs. Averaging across all treatments, the bottom-right panel gives the average beliefs of subjects with hump-shaped beliefs and the average beliefs of all other subjects.

Result 1. The mean and distribution of elicited beliefs—both first-order and second-order—are very similar across all treatments. Most individual subjects have increasing beliefs, but a substantial minority (17%) have hump-shaped beliefs.

³⁵Very few subjects report flat beliefs: only 20 of 350 subjects (5.7%) always report beliefs between 45 and 55%, inclusive.

That beliefs tend to be monotonically increasing means that we expect most deception-averse subjects to send higher messages following heads (Part (i) of Proposition 4), i.e. a positive coin effect. Moreover, monotonic beliefs are not necessary for positive coin effects. Consider the typical pattern for subjects with hump-shaped beliefs: beliefs are minimized at 1 and maximized at 7. A subject with such second-order beliefs who is sufficiently deception-averse will message 1 when tails and 7 when heads—a large positive coin effect. Hence, while such subjects may exhibit negative coin effects, they may only do so if they have partial deception aversion. Moreover, under part (2) of Assumption 1, the only possible negative coin effects for such subjects involve messaging 8 when tails and 7 when heads. Hence, scope for negative coin effects is limited.

3.1 A simple model of boundedly rational beliefs

Our empirical analysis will not make use of any particular model of beliefs. Indeed, a virtue of the theory developed in Section 1 is that it is agnostic about where the beliefs come from. However, the regularities observed in the beliefs data are consistent with a certain family of models, which we offer as a means to better understand the data.

In Appendix 10.2, we show that the beliefs data are well-described by models in which beliefs are anchored on the (perhaps) naive assumption that senders have a tendency to tell the truth. We formalize this using a level k -type model in the spirit of Wang et al. [2010] with level 0-senders that are *truthful* and level 0-receivers that are *credulous*. We show that, due to the positive correlation between coin and die, this model generates strictly increasing second-order beliefs if there is sufficient naïveté. Moreover, it also predicts that sufficiently sophisticated senders will have hump-shaped second-order beliefs that peak at 7. Hence, the model captures the main qualitative patterns in the data.

We also note that, because the sender’s subjective expected payoffs coincide in all games under the *truthful/Bayesian* benchmark (as discussed in Section 2), the model also helps to rationalize why beliefs are so similar across treatments.

4 Aggregate Sender data

Average sender strategies. The left panels of Figure 4 give the average sender strategies for each treatment.

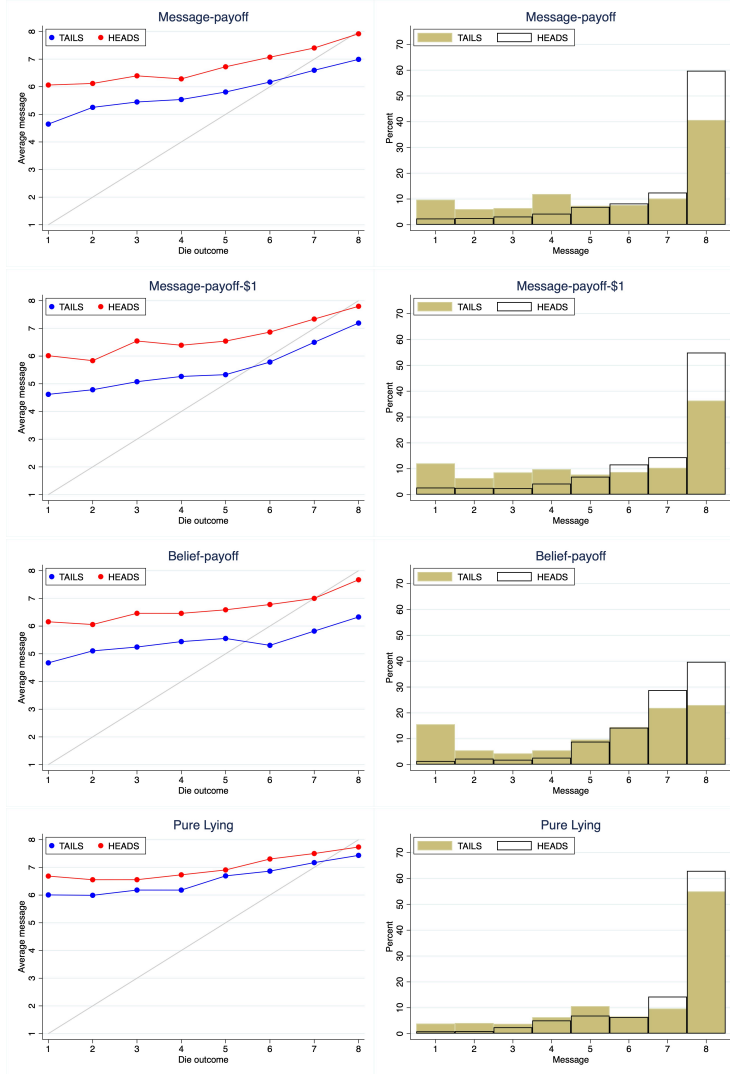


Figure 4: *Average messages conditional on coin and die; distribution of messages conditional on coin.*

There is a clear positive coin effect—the vertical distance between curves—in all deception treatments, consistent with deception aversion and increasing second-order beliefs (Proposition 4). In columns (1)-(4) of Table 3, we quantify these coin effects by regressing the message sent on an indicator for heads and the die outcome. We find average coin effects of 0.94, 1.08, 1.22, and 0.42 in the message-payoff, message-payoff-\$1, belief-payoff, and pure lying treatments, respectively, all of which are highly significant. The coin effects in the deception treatments are more than twice (2.2-2.9 times) the size of those in pure lying, and these differences are highly significant (columns (5)-(7)).

	Coin effect				Difference in coin effects		
	MP (1)	MP-\$1 (2)	BP (3)	PL (4)	MP vs. PL (5)	MP-\$1 vs. PL (6)	BP vs. PL (7)
Heads	0.94*** (0.14)	1.09*** (0.21)	1.22*** (0.25)	0.42** (0.16)	0.42** (0.16)	0.42** (0.16)	0.42** (0.16)
Die	0.29*** (0.03)	0.29*** (0.04)	0.19*** (0.03)	0.20*** (0.05)	0.29*** (0.03)	0.29*** (0.04)	0.19*** (0.03)
Heads $\times$ treatment					0.52** (0.22)	0.67** (0.27)	0.80** (0.29)
Die $\times$ PL					-0.09 (0.05)	-0.10 (0.06)	0.00 (0.05)
PL					1.16** (0.47)	1.44** (0.52)	1.11** (0.48)
Constant	4.51*** (0.26)	4.23*** (0.35)	4.56*** (0.28)	5.67*** (0.39)	4.51*** (0.26)	4.23*** (0.35)	4.56*** (0.28)
Observations	5760	2960	3280	2000	7760	4960	5280

Table 3: *Coin effects: reduced-form evidence of deception aversion.* In columns (1)-(4), for each treatment, we regress the message sent on an indicator for the coin being heads and the die outcome. Columns (1)-(4) correspond to each of the treatments. The coefficient on “Heads” is the average coin effect. In columns (5)-(7), we consider the *difference in* average coin effects between a given treatment and the pure lying treatment. The coefficient on “Heads $\times$ treatment” gives the difference in coin effects between the given treatment and pure lying. Standard errors are clustered at the subject level.

The left panels of Figure 4 show that there is also substantial lying in all treatments. This can be seen by comparing the average messages following each die outcome to the truthful benchmark (upward-sloping gray line). On average, deviations tend to be upward, consistent with lying to increase material payoffs. Despite the significant amount of lying, there are also clear die effects in all treatments (see Table 3), consistent with lying aversion (Proposition 3). In particular, die effects are positive, with the average messages monotonically increasing in the die outcome.

Result 2. In all treatments, the average sender strategies exhibit (1) positive die effects and (2) positive coin effects, with much higher magnitudes in the deception treatments relative to pure lying (2.2-2.9 times higher). The former is consistent with lying aversion; the latter is consistent with deception aversion and increasing second-order beliefs.

Distribution of messages. To better understand which messages senders use, the right panels of Figure 4 plot, for each treatment, histograms of messages conditional on

the outcome of the coin.³⁶

Message-payoff. The two message-payoff treatments (first two right-panels) feature very similar message distributions, suggesting an insensitivity to receiver-payoffs; and hence we pool treatments for purposes of this discussion. *Following heads*, the higher the message, the more often it is sent, with message 8 sent an overwhelming 58% of the time. The other messages 1-7 are sent between 2 and 13% of the time. Hence, subjects strongly favor messaging 8 and rarely ever send the messages 1-4, presumably those that they believe signal that the coin is tails more likely than not. *Following tails*, we see that 8 is still the most common message, being sent 39% of the time. There is thus a 19 percentage point (p.p.) gap in sending 8 following heads versus tails, which we will see plays a crucial role in driving coin effects. Following tails, it is no longer the case that the higher the message, the more often it is sent. Messages 1 and 4 are sent more often than the neighboring messages—and much more often than following heads. This, we will argue, is actually intuitive from a deception-aversion perspective.

Belief-payoff. In belief-payoff (third right-panel), the pattern is similar at the extremes: message 8 is 17 p.p. more likely following heads and message 1 is 14 p.p. more likely following tails. However, while 8 remains the most common message following both heads and tails, much of the mass has been shifted from 8 to 7 and to some extent 6 and 5. What accounts for this downward shift? Consider senders with hump-shaped second-order beliefs. In message-payoff, they maximize expected material payoffs by sending 8; in belief-payoff, they do so by sending lower messages (corresponding to the peak of the hump). Hence, the shift likely results from rational, self-interested behavior.

Pure Lying. In pure lying (bottom-right panel), we see that the entire distribution of messages and the probability of messaging 8 (55-63%) are much less affected by the coin. Unlike in the other treatments, there is no strong tendency to message 1, which is sent only 4% of the time following tails and 1% of the time following heads.

Result 3. (1) In all treatments and conditional on both coin outcomes, 8 is the most common message. (2) In the deception treatments (but not pure lying), the distribution of messages is qualitatively different following heads versus tails, with 1 sent a significant

³⁶Note that separation of the two conditional message distributions does not imply coin effects because (1) coin effects are defined conditional on die outcome and (2) the two conditional die distributions are different. For example, a truthful sender implies separation, but does not exhibit coin effects.

fraction of the time following tails (but not heads).

Learning? Since there is no feedback, there can be no conventional learning, but we analyze later rounds to rule out that erratic initial play or exploration drives the observed effects. In Online Appendix 10.1, we replicate this section’s analysis using only the last 20 rounds. The results are very similar, but with slightly larger average coin effects in the deception treatments (1.00, 1.29, and 1.23) and smaller in pure lying (0.20). If anything, the gap between deception and pure lying treatments grows over time.

5 Heterogeneity in individual sender strategies

Because coin effects indicate deception aversion and die effects indicate lying aversion, we are able to summarize heterogeneity through two reduced-form statistics.

First, we consider the *average coin effect* for individual i :

$$C_i := \frac{\sum_{d'} \mathbf{1}\{i \text{ observes both } (1, d') \text{ and } (0, d')\} (\bar{\sigma}_S^i(1, d') - \bar{\sigma}_S^i(0, d'))}{|\{d' | i \text{ observes both } (1, d') \text{ and } (0, d')\}|},$$

where $\bar{\sigma}_S^i(\theta, d')$ is the average message following coin-die realization (θ, d') and $\bar{\sigma}_S^i(1, d') - \bar{\sigma}_S^i(0, d')$ is the *coin effect conditional on d'* . Hence, C_i is the equal-weighted average of conditional coin effects over all d' for which both $(1, d')$ and $(0, d')$ are observed.

Second, we consider the *average lie* for individual i :

$$L_i := \frac{\sum_{d'} \mathbf{1}\{i \text{ observes } d'\} (\bar{\sigma}_S^i(d') - d')}{|\{d' | i \text{ observes } d'\}|},$$

where $\bar{\sigma}_S^i(d')$ is the average message conditional on d' .

To better understand these statistics, consider the statistics implied by a few benchmark strategies.³⁷ The *truthful* strategy, employed by a sender with very strong lying aversion, yields $(L_i, C_i) = (0, 0)$. The *always-8* strategy, which is the self-interested

³⁷These computations assume complete data, i.e., that every coin-die outcome is realized at least once. However, some coin-die realizations may be missing due to randomness, which may slightly bias the statistics. Nevertheless, we confirm that this is of no concern in our data. With 40 observations for each subject, more than 90% of subjects observe at least 14 of 16 possible coin-die outcomes, and all results are virtually unchanged if we linearly interpolate/extrapolate the missing observations before conducting the analysis. We note also that C_i is completely unaffected by incompleteness unless the strategy features *both* coin and die effects, in which case there may or may not be some bias.

or material payoff-maximizing strategy in the message-payoff game and in the belief-payoff game under increasing second-order beliefs, yields $(L_i, C_i) = (3.5, 0)$. *Always-7* yields $(L_i, C_i) = (2.5, 0)$. This may be used by self-interested senders in the belief-payoff game if they have hump-shaped second-order beliefs. The strategy of messaging 8 following heads and 1 following tails, which we refer to as $(8,1)$, yields $(L_i, C_i) = (0, 7)$. This is the strategy of an infinitely deception-averse sender with increasing second-order beliefs. More generally, the family of strategies $(8, x)$ for $x \in \{1, \dots, 8\}$ yields $(L_i, C_i) = (\frac{1}{2}x - \frac{1}{2}, 8 - x)$. These strategies are used by purely deception-averse senders with varying degrees of deception aversion and increasing second-order beliefs. For fixed second-order beliefs, lower x implies greater deception aversion (Corollary 2).

Both C_i and L_i are signed. A subject for whom $C_i > 0$ ($C_i < 0$) exhibits a positive (negative) coin effect on average, and a subject for whom $L_i > 0$ ($L_i < 0$) tends to lie upward (downward) on average. Note that upward and downward lies cancel out, e.g. $(8,1)$ implies $L_i = 0$ even though most messages are lies. Because C_i and L_i are signed, random behavior will not systematically yield any particular sign. Conversely, if they do tend to have a particular sign, this cannot be an artifact of unbiased noise.

Figure 5 plots (L_i, C_i) for all subjects in each of the treatments, with the averages displayed in the top-right of each panel.³⁸ There is clearly substantial heterogeneity in all treatments, as well as significant regularities. In the deception treatments, many subjects have positive coin effects, including some with very large effects (e.g. 22% of subjects with $C_i > 2$). On the other hand, the average positive coin effect in pure lying is entirely driven by two outlying subjects. Negative coin effects are rare in all treatments.

Inspecting Figure 5 more closely, there are clear clusters of truthful and always-8 strategies in all treatments. In pure lying, no other strong clusters appear. In the deception treatments, a good number of subjects use a strategy that is approximately $(8, x)$ for different values of $x < 8$, in particular $(8,7)$, $(8,4)$, and $(8,1)$. In belief-payoff, there is a cluster of always-7, of similar magnitude as always-8 (consistent with the finding of hump-shaped beliefs). There are also many subjects with clear positive coin effects, but who do not conform exactly to these simple strategies. Their strategies feature both coin and die effects, suggesting a mix of both deception and lying aversion.

³⁸The average C_i ranges from 0.96 to 1.15 in the deception treatments, compared to 0.34 in pure lying—a 2.8-3.3-fold increase. This is a slightly larger difference than that estimated in Section 4, due to a different aggregation (a representative agent approach vs. averaging individual effects).

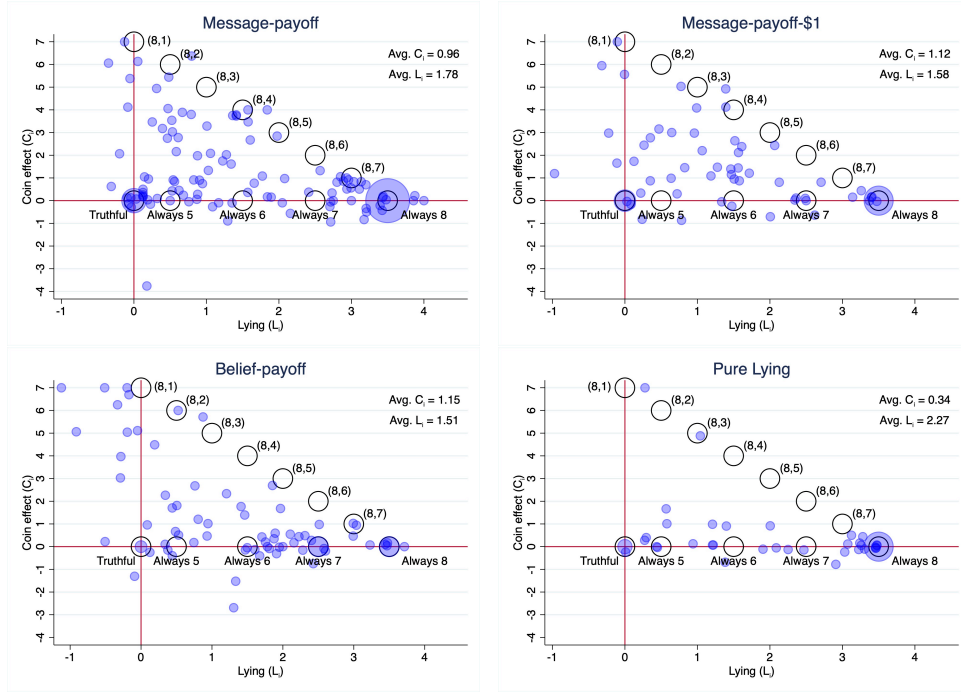


Figure 5: *Heterogeneity in individual sender strategies.*

Result 4. (1) All treatments show significant heterogeneity in individual sender strategies, with clusters of *truthful* and *always-8*. (2) In the deception treatments (but not pure lying), many subjects use strategies that involve positive coin effects, consistent with deception aversion and increasing second-order beliefs.

6 Potential Confounds and Robustness

6.1 Experimenter demand or entitlement effects?

One possible concern is that subjects would exhibit coin effects even if the coin were irrelevant to payoffs, which could be due to any number of reasons unrelated to deception aversion, e.g. experimenter demand or entitlement effects.

In the pure lying game, the coin is *not* payoff-relevant. Therefore, there is no opportunity to deceive, or even to expect the receiver to form a belief about the coin at all. Hence, even a deception-averse subject should not exhibit coin effects. However, experimental subjects, being exposed to seemingly irrelevant information, may try to use

it for any number of reasons. For example, the very inclusion of the coin may make subjects question its relevance, leading them to condition their behavior on its outcome—a form of *experimenter demand*. Another possibility is that, since higher die outcomes are more likely following heads, some subjects may feel “entitled” to higher payment and therefore send higher messages following heads—a form of *entitlement effect* (Hoffman et al. [1994]). Hence, the pure lying game serves as a control to determine if coin effects are the result of genuine deception aversion as opposed to such alternative factors.

In Figure 6, we plot the CDFs of individual coin effects in the message-payoff and pure lying treatments, for which the sender has exactly the same material payoff function. We find that the coin effects in message-payoff stochastically dominate those in pure lying, with much larger effects across the entire distribution of subjects. This suggests that coin effects in the deception games, consistent with deception aversion, cannot be explained by alternative “irrelevant” factors.

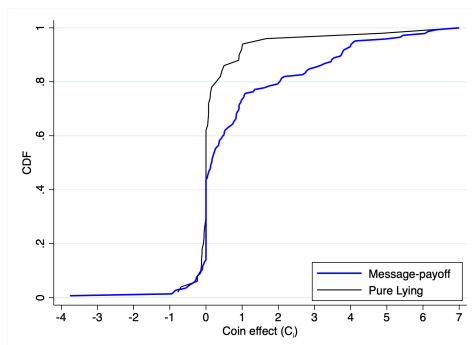


Figure 6: *Distribution of individual coin effects: message-payoff vs. pure lying.*

6.2 Alternative sender preferences?

In practice, the senders’ expected material payoffs in the message-payoff and belief-payoff games are plausibly similar (as they exactly coincide in the truthful/Bayesian benchmark). However, the two games have a qualitatively different character, which may affect attitudes toward deception. In message-payoff, the sender’s payoff depends only on her message, and so deception comes purely as an *externality* of individually rational behavior. In belief-payoff, there is a *direct incentive* to deceive.

A natural conjecture is that this direct incentive would decrease deception aversion, perhaps by weakening any identity-based concerns (cf. Bénabou and Tirole [2011]) that

may underlie it. On the other hand, the direct incentive may increase deception aversion if it makes deception opportunities more salient. However, we show in Figure 7 that the two treatments actually result in a very similar distribution of individual coin effects. As the two games embody different extreme cases of sender preferences, this suggests that deception aversion may be robust across a variety of strategic settings.

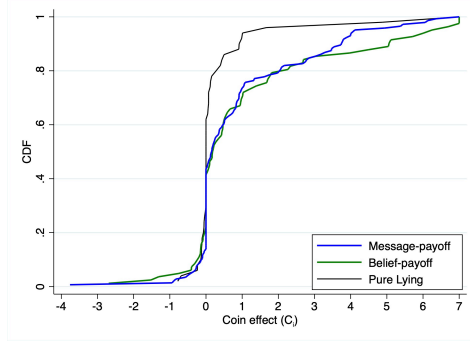


Figure 7: *Distribution of individual coin effects: message-payoff vs. belief-payoff.*

6.3 Social preferences?

By engaging in deception, the sender believes she is inducing inaccurate beliefs in the receiver, which in turn has negative payoff consequences for the receiver. An important question therefore is whether the effects we observe are due to deception aversion per se, or if they are fully driven by *social* or *other-regarding* preferences over final outcomes.

To understand the issue, we generalize utility function (1) for the deception games to explicitly allow for social preferences:

$$\tilde{\phi}^g(m, \theta, d; \beta) = \tilde{u}(p_S^g(m; \beta), \gamma p_R^g(m, \theta; \beta)) - c^{lie}(m, d) - c^{dec}(\beta(m), \theta; \gamma). \quad (2)$$

We have simply replaced self-interested material utility $u^g(m; \beta)$ with $\tilde{u}(x, \gamma y)$, where $x = p_S^g(m; \beta)$ and $\gamma y = \gamma p_R^g(m, \theta; \beta)$ are the material payoffs that the sender expects to induce in the sender and receiver, respectively, in game $g \in \{M, B\}$.³⁹ Parameter $\gamma \in \{1, 0.1\}$ is the “scale” of receiver payoffs (relative to the sender) that spans the baseline message-payoff game ($\gamma = 1$) and message-payoff-\$1 ($\gamma = 0.1$) in which the

³⁹ $p_S^M(m) = 0.32 + 0.04m$, $p_S^B(m; \beta) = \beta(m)$, and, for both $g \in \{M, B\}$: $p_R^g(m, 0; \beta) = 1 - \beta(m)^2$ and $p_R^g(m, 1; \beta) = 1 - (1 - \beta(m))^2$. The expressions for $p_R^g(m, \theta; \beta)$ are based on the binarized scoring rule.

receiver's value of holding accurate beliefs is 10 times lower. By allowing γ to enter $c^{dec}(\cdot)$, we now make explicit that the cost of deception can depend directly on its payoff consequences⁴⁰ and that this is distinct from social preferences which act through $\tilde{u}(\cdot)$.

We note that, since θ enters $\tilde{u}(\cdot)$ directly because it is payoff-relevant to the receiver, social preferences (e.g. altruism and/or inequality aversion) can give rise to coin effects even without deception costs. However, we offer *three arguments* to suggest that social preferences cannot fully explain the observed effects.

Dictator games. In half of the message-payoff sessions (those in which second-order beliefs are elicited) and all of the message-payoff-\$1 sessions ($n = 148$ total), the questionnaire included an incentivized dictator game.⁴¹ In our variant, subjects were randomly matched in pairs, with both subjects in the pair choosing an allocation as if they were the dictator. One subject in the session was then randomly chosen, and her choice as dictator was implemented (with her matched subject as the recipient). Dictators chose to give $y \in \{0, 10, \dots, 100\}$ to the recipient and keep $100 - y$ for themselves. Hence, subject i 's choice of y_i gives a measure of her degree of altruism. For comparability to the main task, payoffs were in units of probability points of earning \$10 (independently determined across dictator and recipient).

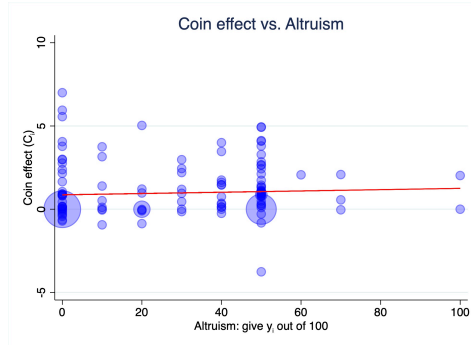


Figure 8: *Coin effects are uncorrelated with measures of altruism.*

In Figure 8, we give a scatter plot of individual coin effects C_i against altruism y_i . We find that there is *no correlation* between the two, with an estimated correlation

⁴⁰For example, if $c^{dec}(\beta(m), \theta; \gamma) = f(|\beta(m) - \theta|; \gamma)$ for some increasing f , then deception costs are increasing in (1) the size of deception and (2) even more so to the extent it hurts the receiver.

⁴¹These sessions were run after our original sessions in response to referee comments. The inclusion of the dictator game was also suggested by referees.

coefficient of 0.06 ($p = 0.46$). This suggests that the observed coin effects may not be driven by social preferences over final outcomes. Of course, as with all correlational evidence, this should be interpreted cautiously.

Insensitivity to changes in receiver payoffs. Consider a standard dictator game experiment. The dictator must choose an allocation for herself (x) and the recipient (y), with a tradeoff determined by a budget constraint $x + py = M$. It has been observed many times (e.g., Jakiela [2013]), that when the price p of allocating to the recipient increases, there is a tendency for dictators to choose a larger allocation for themselves (higher x) at the expense of the recipient (lower y). This suggests regard for both one’s own payoff and that of the recipient—i.e. altruism. It is also compatible with some degree of inequality aversion (but not extreme forms that predict the dictator chooses *lower* x as p increases for the sake of equality).

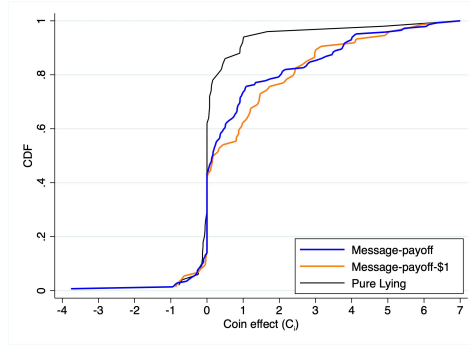


Figure 9: *Distribution of individual coin effects: message-payoff vs. message-payoff-\$1.*

Based on this, if coin effects are driven largely by social preferences, one would expect that reducing the payoff consequences to the receiver, as in the comparison between message-payoff ($\gamma = 1$) and message-payoff-\$1 ($\gamma = 0.1$), would reduce coin effects. To illustrate, consider the simple case in which there are no lying or deception costs and second-order beliefs are strictly increasing. Let us suppose further that, *following heads*, $\tilde{u}(\cdot)$ is such that senders will choose $m = 8$ for all values of γ as doing so maximizes both sender and receiver payoffs. *Following tails*, however, the tradeoff resembles that in dictator games: higher messages increase the sender’s expected payoff (higher x) at the expense of the receiver (lower γy). When γ is lower, choosing higher messages continues to be equally beneficial to the sender while having a smaller negative effect

on the receiver: it is as if the “price” of y increases, exactly as in standard dictator games. Hence, when γ is lower, we should expect higher messages following tails—and therefore reduced coin effects. However, as shown in Figure 9, we find that both message-payoff treatments result in a very similar distribution of individual coin effects, which we interpret as suggesting a minimal role of social preferences in driving coin effects.

Pure lying as a control. Suppose that we could compare behavior in the deception games to a *control game* in which messages lead to the same sender-receiver allocations and incur the same lying costs, but there is no possibility to deceive—and therefore no deception costs. In other words, suppose the control game induces utility function

$$\tilde{\phi}^0(m, \theta, d) = \tilde{u}(p_S^0(m), p_R^0(m, \theta)) - c^{0,lie}(m, d),$$

where $(p_S^g(m; \beta), p_R^g(m, \theta; \beta)) \approx (p_S^0(m), p_R^0(m, \theta))$ for all (m, θ) and $c^{0,lie}(m, d) \approx c^{lie}(m, d)$ for all (m, d) .⁴² In such case, the utility difference $\tilde{\phi}^0(m, \theta, d) - \tilde{\phi}^g(m, \theta, d; \beta)$ would approximately equal $c^{dec}(\beta(m), \theta)$, and therefore any difference in behavior between deception and control games would be fully attributable to deception costs. Note that, because $\tilde{u}(\cdot)$ drops out, this claim allows for arbitrary social preferences.

We argue that, *conditioning on tails* ($\theta = 0$), the pure lying game makes a good control for the deception games. Table 4 shows the material payoffs the sender expects to induce in both sender and receiver as a function of the message m , given second-order beliefs β and that the coin is tails. Note that the receiver’s payoff of $1 - \beta(m)^2$ in the deception games is based on the binarized scoring rule.

Game	Sender’s payoff (p_S^g)	Receiver’s payoff (p_R^g)
Message-payoff	$0.32 + 0.04m$	$1 - \beta(m)^2$
Belief-payoff	$\beta(m)$	$1 - \beta(m)^2$
Pure lying	$0.32 + 0.04m$	$0.68 - 0.04m$

Table 4: *Sender’s expected material payoffs for both sender and receiver conditional on tails.*

If β is strictly increasing, all games feature a similar *qualitative* tradeoff between sender and receiver material payoffs, with lower messages being more generous to the receiver at the expense of the sender. The *quantitative* tradeoffs are also plausibly similar

⁴²We focus on the base message-payoff game ($\gamma = 1$) and belief-payoff game for which receiver payoffs are on the same scale as in the pure lying game. Hence, we drop γ from the notation.

across all games, though this is sensitive to β . In the case that $\beta(m) = 0.32 + 0.04m$ (the truthful/Bayesian benchmark), sender payoffs are identical across all games. Moreover, the marginal changes in receiver payoffs as a function of the message are also similar across games: $(1 - \beta(m)^2) - (1 - \beta(m+1)^2) \approx 0.04$ for $m \in \{1, 2, \dots, 7\}$.⁴³

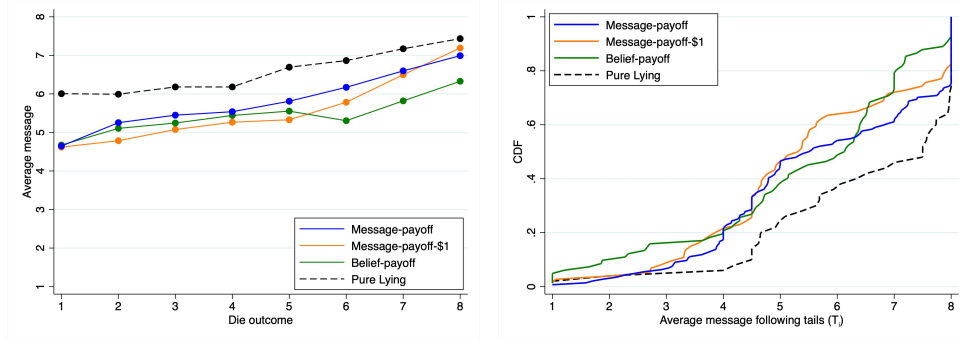


Figure 10: *Average messages conditional on tails.*

Under Assumption 1, following tails, lower messages in the deception games imply lower deception costs. Hence, within our framework and assuming the validity of the control, if the sender sends lower messages following tails in the deception games (relative to the control), this can only be explained by deception aversion. Intuitively, following a given coin-die realization $(\theta = 0, d)$, the sender’s message in the pure lying game reveals her optimal sender-receiver allocation in the absence of deception costs. If the sender chooses lower messages in the deception games following the same coin-die realization, this can only mean that she is changing her behavior to avoid deception costs.

In Figure 10, we plot the average messages sent after each die outcome, conditional on tails, in all treatments. We find that, compared to pure lying, average messages following tails are much lower (and statistically so) in the deception treatments (left panel). To confirm this is not driven by a handful of subjects, we also compute the average message following tails for each individual i ⁴⁴ and show that the distribution of individual effects in pure lying dominates that in all other treatments (right panel). Hence, there is a

⁴³The exact values range from 0.03 to 0.05. The “level” of receiver payoffs is higher in pure lying than in the deception games, but all standard social preference models predict that this makes the sender less willing to sacrifice her payoffs to benefit the receiver, making our analysis even more conservative.

⁴⁴The average message following tails for individual i is $T_i = \frac{\sum_{d'} \mathbf{1}\{i \text{ observes } (0, d')\}(\bar{\sigma}_S^i(0, d'))}{|\{d | i \text{ observes } (0, d)\}|}$, where $\bar{\sigma}_S^i(0, d')$ is the average message conditional on $(0, d')$.

much stronger tendency to send lower messages following tails in the deception games, which can only be explained by deception aversion within our framework.

Result 5. The coin effects observed in the deception games cannot be fully explained by experimenter demand, entitlement effects, or social preferences, and are robust to alternative sender preferences.

7 Structural model and estimates

We develop a simple structural model, which we fit to the data of individual sender-subjects. While relying on additional structural assumptions, this allows for a statistical determination as to which subjects are significantly averse to deception, lying, both, or neither—a classification that cannot be driven by noise.

Our structural model is based off of utility function (1). In our baseline specification, we assume that both lying and deception costs are linear:

$$\phi^g(m, \theta, d; \kappa^{lie}, \kappa^{dec}, \beta) = u^g(m; \beta) - \kappa^{lie}|m - d|/7 - \kappa^{dec}|\beta(m) - \theta|, \quad (3)$$

where $\kappa^{lie} \geq 0$ and $\kappa^{dec} \geq 0$ are the lying and deception cost parameters, respectively. Note that both lying and deception costs are based on *absolute* distances.

In addition to the two cost parameters, which we will estimate, we must account for the unobserved second-order beliefs. We take an empirical approach, using the elicited belief data. Because individual-level elicited beliefs are quite noisy for many subjects, we will take averages across subjects. However, the simple average across all subjects does not allow for any heterogeneity across individuals nor does it capture the fact that there are two distinct groups of subjects, those with monotonic beliefs and those with hump-shaped beliefs that peak at 7. Hence, we take two averages: the average beliefs of subjects with hump-shaped beliefs and the average beliefs of all other subjects as depicted in the bottom-right panel of Figure 3. We then consider the family of belief functions $\beta_\alpha(\cdot)$ depicted in Figure 11: an α -weighted average of the two average belief functions, with α being the weight on the hump-shaped function. This family of belief functions is only one parameter, captures the important qualitative features in the data, and is closely related to the level k model of Online Appendix 10.2.

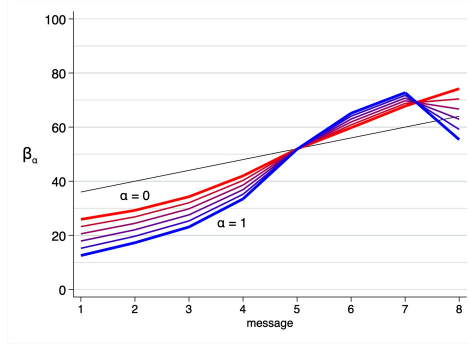


Figure 11: *Parametric family of second-order beliefs (β_α) based on the data. $\beta_{\alpha=1}$ corresponds to the average beliefs of subjects with hump-shaped beliefs, $\beta_{\alpha=0}$ corresponds to the average beliefs of all other subjects, and $\alpha \in [0, 1]$ interpolates between the two functions.*

Hence, we consider the utility function $\phi^g(m, \theta, d; \kappa^{lie}, \kappa^{dec}, \beta_\alpha)$ and assume a logit error structure with noise parameter $\lambda \geq 0$. For each individual i , we then estimate the four parameters $(\kappa^{lie}, \kappa^{dec}, \alpha, \lambda)$ by maximizing the log-likelihood function

$$\mathcal{L}(\kappa^{lie}, \kappa^{dec}, \alpha, \lambda) = \sum_{t=1}^{40} \ln \left(\frac{\exp(\lambda \cdot \phi^g(m_t, \theta_t, d_t; \kappa^{lie}, \kappa^{dec}, \beta_\alpha))}{\sum_{m'=1}^8 \exp(\lambda \cdot \phi^g(m', \theta_t, d_t; \kappa^{lie}, \kappa^{dec}, \beta_\alpha))} \right),$$

where, for individual i in round $t \in \{1, \dots, 40\}$, $(\theta_t, d_t) \in \{0, 1\} \times \{1, \dots, 8\}$ is the coin-die realization and $m_t \in \{1, \dots, 8\}$ is the message sent.

To determine which of κ^{lie} and κ^{dec} add explanatory power, we consider the full model $M_1 = (\kappa^{lie}, \kappa^{dec}, \alpha, \lambda)$ as well as three restricted models: $M_2 = (\kappa^{lie} = 0, \kappa^{dec}, \alpha, \lambda)$, $M_3 = (\kappa^{lie}, \kappa^{dec} = 0, \alpha, \lambda)$, and $M_4 = (\kappa^{lie} = 0, \kappa^{dec} = 0, \alpha, \lambda)$. We then perform standard likelihood ratio (LR) tests to determine the best-fitting model, penalizing models with more parameters.⁴⁵ If M_1 ($\kappa^{lie} > 0$ and $\kappa^{dec} > 0$) is the best model, we say that the individual is averse to both lying and deception. If M_2 ($\kappa^{lie} = 0$ and $\kappa^{dec} > 0$) is the best model, we say that the individual is only averse to deception. If M_3 ($\kappa^{lie} > 0$ and $\kappa^{dec} = 0$) is the best model, we say that the individual is only averse to lying. Otherwise, M_4 ($\kappa^{lie} = \kappa^{dec} = 0$) is the best model, and we conclude that the individual is averse to neither lying nor deception. Importantly, by using statistical tests, if a subject is

⁴⁵If $\mathcal{L}(M_1) > \max\{\mathcal{L}(M_2), \mathcal{L}(M_3)\} + 3.841$ and $\mathcal{L}(M_1) > \mathcal{L}(M_4) + 5.991$, then M_1 is the best model. If $\mathcal{L}(M_2) > \mathcal{L}(M_3)$, $\mathcal{L}(M_1) < \mathcal{L}(M_2) + 3.841$, and $\mathcal{L}(M_2) > \mathcal{L}(M_4) + 3.841$, then M_2 is the best model. If $\mathcal{L}(M_3) > \mathcal{L}(M_2)$, $\mathcal{L}(M_1) < \mathcal{L}(M_3) + 3.841$, and $\mathcal{L}(M_3) > \mathcal{L}(M_4) + 3.841$, then M_3 is the best model. Otherwise, M_4 is the best model.

classified as deception-averse (say), they are significantly so.

Game	Averse to:			
	Lying and deception ($\kappa^{lie} > 0, \kappa^{dec} > 0$)	Deception only ($\kappa^{lie} = 0, \kappa^{dec} > 0$)	Lying only ($\kappa^{lie} > 0, \kappa^{dec} = 0$)	Neither ($\kappa^{lie} = \kappa^{dec} = 0$)
Message-payoff	17% (25)	13% (19)	35% (50)	35% (50)
Message-payoff-\$1	24% (18)	9% (7)	36% (27)	30% (22)
Belief-payoff	15% (12)	12% (10)	40% (33)	33% (27)
Pure lying	2% (1)	6% (3)	34% (17)	58% (29)

Table 5: *Summary of structural estimates.* This table summarizes the percentage of subjects (number of subjects in parenthesis) in each treatment who are best described by each model.

In Table 5, we give the percentage and number of subjects in each treatment who are best explained by each of the models. We do find slightly less deception aversion in belief-payoff compared to message-payoff, but the general patterns are similar: an incentive to deceive does *not* drive out deception aversion. Pooling across the deception treatments, we find that 18% of subjects are averse to both lying and deception, 12% to deception only, 37% to lying only, and 33% to neither. Hence, 30% of subjects are averse to deception.

As a placebo test, we fit the same model to the individual subject data in the pure lying treatment where there is no opportunity to deceive. Reassuringly, we find that only 8% (4 out of 50) are classified as deception-averse, a much smaller percentage than in the other treatments. We find that a similar percentage of subjects are purely lying-averse (34%) and a much higher percentage are neither lying- nor deception-averse (58%).

Robustness. In Online Appendix 10.4, we show that the results are robust to alternative structural models that vary in terms of the family of second-order beliefs considered as well as the shape of the lying and deception cost functions.⁴⁶

Result 6. In the deception treatments, the structural model implies that 18% of subjects are averse to both lying and deception, 12% to deception only, and 37% to lying only. These results are similar across variations of the game and robust to alternative modeling assumptions.

⁴⁶Across the alternative models, we assume that an individual subject’s second-order beliefs are either (1) her own elicited beliefs (in which case first-order beliefs proxy for second-order for some subjects), (2) the simple average of all elicited beliefs, or (3) the beliefs implied by the truthful/Bayesian benchmark. We assume that the cost functions are either (1) linear or (2) quadratic.

7.1 Individual subjects

To better visualize individual subject behavior and understand the implications of model (3), we plot the average messages, conditional on coin and die, for individual subjects, superimposed with predictions of the best-fitting model.

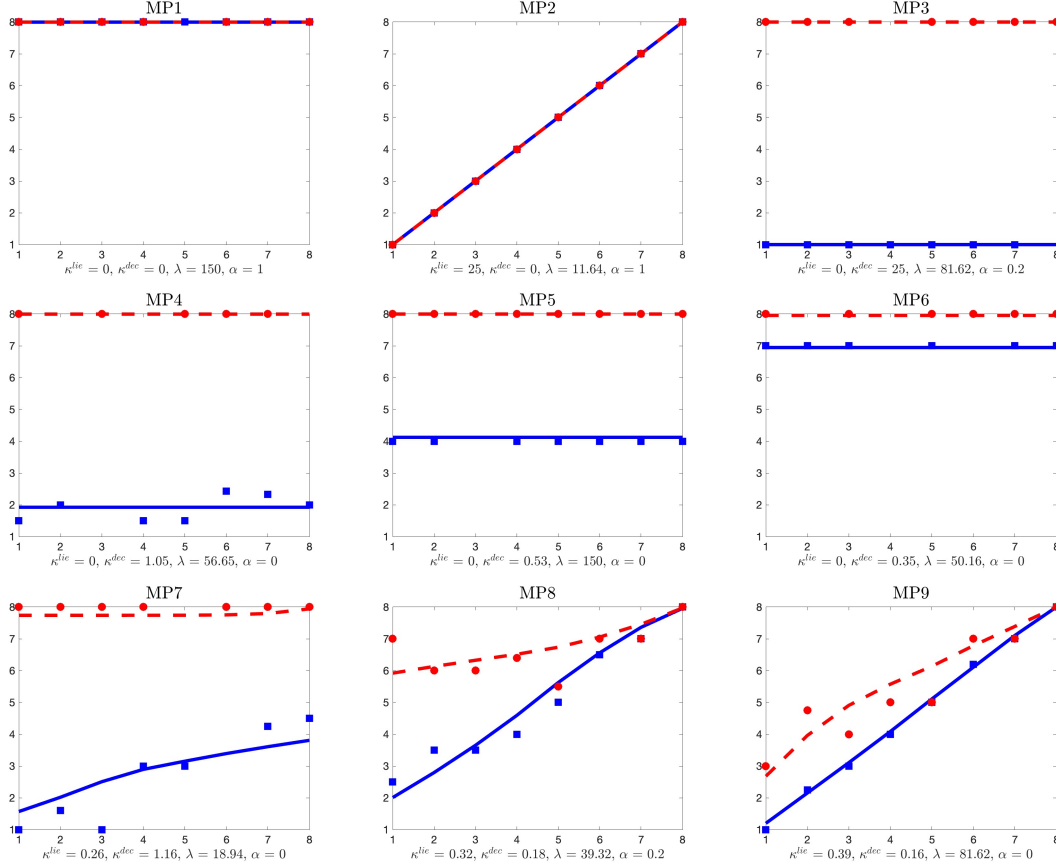


Figure 12: *Individual subjects (message-payoff).* Each panel gives an individual subject's average messages conditional on coin and die outcome, superimposed with the average messages predicted by the best-fitting model. Parameter estimates are given at the bottom of each panel.

Message-payoff. Figure 12 gives plots for nine subjects in the message-payoff treatment, who are all well-captured by the model. Subject MP1 is classified as neither lying- nor deception-averse and therefore always messages 8 (always-8). MP2 is extremely lying-averse, and therefore always tells the truth (truthful). MP3 is extremely deception-averse, always messaging 8 when heads and 1 when tails ((8,1)). MP4 is a bit less deception-averse, always messaging 8 when heads and 2 when tails ((8,2)). MP5 is

less deception-averse yet, always messaging 8 when heads and 4 when tails $((8,4))$. This behavior, which we find in 6 subjects in the message-payoff treatment, is particularly interesting. We conjecture that, for these subjects, a message of 4 maximizes induced second-order beliefs among induced beliefs that favor tails. Hence, they are maximizing their payment, subject to not deceiving the receiver so much that their beliefs go “in the wrong direction.” MP6 is only slightly deception-averse, always messaging 8 when heads and 7 when tails $((8,7))$.

The remaining subjects, MP7-9, who exhibit both coin and die effects, are classified as both deception- and lying-averse. They are ordered, from left to right, in order of decreasing deception aversion. It is interesting that the model can capture these rich patterns, including cases like MP7 where the sender’s strategy involves very different “slopes” following heads versus tails.

Belief-payoff. Figure 13 gives plots for nine subjects in the belief-payoff treatment, who are also well-captured by the model. We do observe subjects who are seemingly neither averse to lying nor deception (always-8), extremely lying-averse (truthful), or extremely deception-averse with increasing second-order beliefs $((8,1))$, but such plots look identical to those in the message-payoff treatment, so we omit them here.

Referring to Figure 13, subject BP1 is classified as neither averse to lying nor deception, but is estimated to have hump-shaped second-order beliefs ($\alpha = 1$). She always messages 7 (always-7), which she believes maximizes the receiver’s belief and thus her own payment. BP2 is also neither averse to lying nor deception, but tends to message 6 (always-6). This type of subject is not perfectly captured by the model as it does not allow for second-order beliefs that peak at 6. BP3 is extremely deception-averse, but with hump-shaped second-order beliefs ($\alpha = 1$). As such, she always messages 7 when heads and 1 when tails $((7,1))$. The remaining subjects, BP4-9, are both lying- and deception-averse, arranged in order of decreasing deception aversion.

Unexplained patterns. The structural model provides a good description of the data for many subjects, but it does not capture the complete richness. Some subjects exhibit patterns that cannot be captured qualitatively by the model, but nevertheless suggest regularities (e.g. alternative cost functions or “deceptive heuristics”). In Online Appendix 10.3, we discuss these patterns in detail.

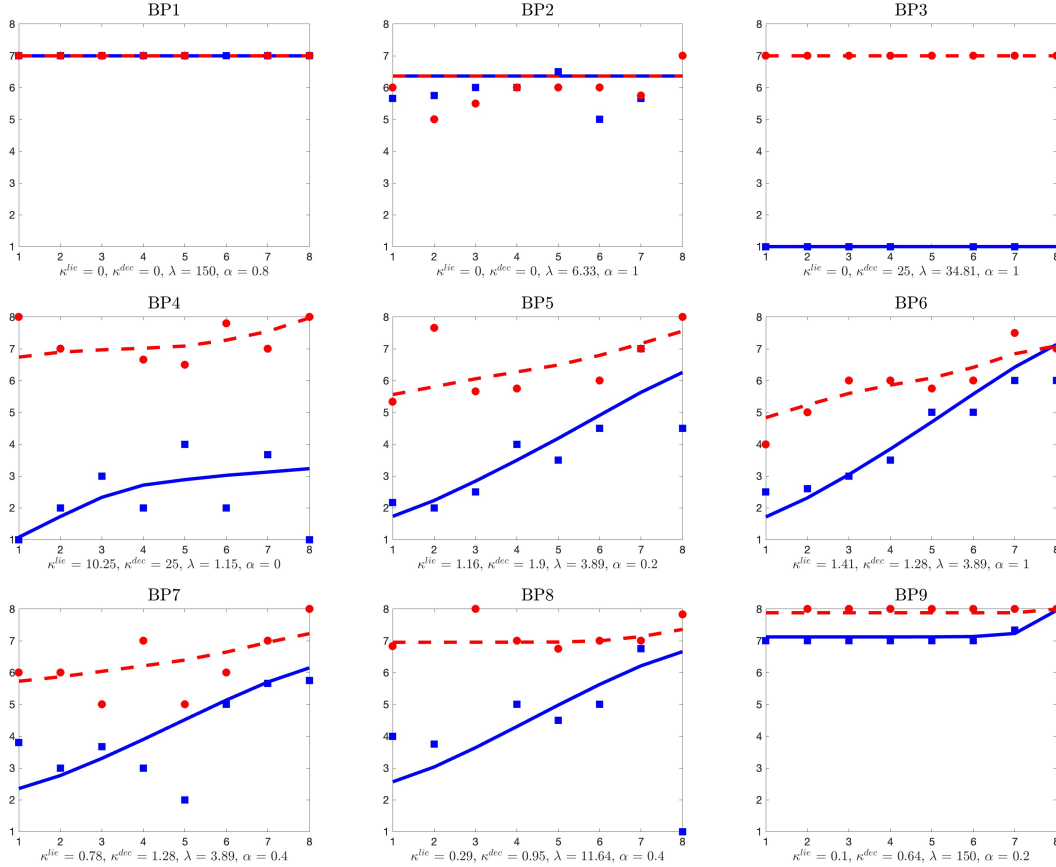


Figure 13: *Individual subjects (belief-payoff)*. Each panel gives an individual subject’s average messages conditional on coin and die outcome, superimposed with the average messages predicted by the best-fitting model. Parameter estimates are given at the bottom of each panel.

8 Conclusion

Communicating private information is central to almost every economic and social interaction. In many cases, there are opportunities both to *lie* by misreporting the truth and to *deceive* by inducing inaccurate beliefs about some payoff-relevant state. While a large and influential literature has argued for lying aversion, there is very little existing evidence for aversion to deceiving others, which may be just as empirically relevant.

In part, the lack of studies documenting deception aversion is unsurprising as the distinction between lying and deception has only recently been formalized (Sobel [2020]). Perhaps the bigger issue, however, is that deception depends on second-order beliefs, a complex and fundamentally unobservable object. This makes studying deception difficult

without making strong auxiliary assumptions or eliciting second-order beliefs directly. In this paper, we take a different approach, introducing a simple game and theoretical framework that allows us to identify deception aversion—and separate it from lying aversion—with minimal assumptions on second-order beliefs. We find strong evidence for deception aversion that cannot be explained by social preferences and is robust to varying strategic features of the game.

Our work is an early step toward a systematic study of deception aversion and suggests many directions for future work. In particular, the “coin effect,” which captures an individual’s degree of deception aversion, may find broad application. For instance, it can be used to study the relationship between deception aversion and payoff consequences, paralleling one of the major themes within the lying aversion literature. This would help to refine models of deception costs, informing theoretical applications. It would also be valuable to examine lying and deception aversion across different populations and cultures, where norms governing honesty may substantially differ. More broadly, because our approach identifies deception aversion from choice data alone—without assuming equilibrium or eliciting beliefs—it may be adapted to study deception in the field or using observational data. Beyond deception, the approach may prove useful for identifying other belief-dependent preferences, even when the relevant beliefs are not directly observed.

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9 Appendix

9.1 Omitted proofs

Proof of Proposition 1. In message-payoff, messaging T is strictly dominant for any (θ, d) , so it must be that $\sigma_S(T|\theta, d) = 1$ in any PBE. Hence, observing message T , the receiver will not update beliefs. In belief-payoff, suppose there is a message m' sent with positive probability in equilibrium after which the receiver forms beliefs $b' \neq p_\Theta(1)$ and therefore reports $b = b'$. Without loss, we may take $b' > p_\Theta(1)$, implying the existence of another message m'' sent with positive probability in equilibrium after which the receiver forms belief $b'' < p_\Theta(1)$ and therefore reports belief $b = b''$. But this cannot be an equilibrium because it is strictly better for the sender to message m' than m'' . Hence,

there can be no belief-updating in equilibrium, which only occurs if $(\theta = 0)$ -senders and $(\theta = 1)$ -senders use strategies that induce the same distribution over messages. $\square$

Proof of Proposition 3. Suppose for purposes of contradiction that $c^{lie}(\sigma_S^*(\theta, d), d') \leq c^{lie}(\sigma_S^*(\theta, d), d)$ and $c^{lie}(\sigma_S^*(\theta, d'), d) \leq c^{lie}(\sigma_S^*(\theta, d'), d')$. By definition of $\phi^g(\cdot)$, we have that $c^{lie}(\sigma_S^*(\theta, d), d') \leq c^{lie}(\sigma_S^*(\theta, d), d) \implies \phi^g(\sigma_S^*(\theta, d), \theta, d') \geq \phi^g(\sigma_S^*(\theta, d), \theta, d)$. By definition of $\sigma_S^*(\cdot)$, we have that $\phi^g(\sigma_S^*(\theta, d'), \theta, d') > \phi^g(\sigma_S^*(\theta, d), \theta, d')$ and thus that $\phi^g(\sigma_S^*(\theta, d'), \theta, d') > \phi^g(\sigma_S^*(\theta, d), \theta, d)$. A symmetric argument yields $\phi^g(\sigma_S^*(\theta, d), \theta, d) > \phi^g(\sigma_S^*(\theta, d'), \theta, d')$, a contradiction. $\square$

Proof of Proposition 4. (i): For purposes of contradiction, suppose $\sigma_S^*(1, d) < \sigma_S^*(0, d)$, which implies by Assumption 1 that $0 \leq \beta(\sigma_S^*(1, d)) \leq \beta(\sigma_S^*(0, d)) \leq 1$ and thus that (a) $c^{dec}(\beta(\sigma_S^*(0, d)), 0) \geq c^{dec}(\beta(\sigma_S^*(1, d)), 0)$ and (b) $c^{dec}(\beta(\sigma_S^*(1, d)), 1) \geq c^{dec}(\beta(\sigma_S^*(0, d)), 1)$. If $u^g(\sigma_S^*(1, d)) - c^{lie}(\sigma_S^*(1, d), d) \geq u^g(\sigma_S^*(0, d)) - c^{lie}(\sigma_S^*(0, d), d)$, then (a) implies $\phi^g(\sigma_S^*(1, d), 0, d) \geq \phi^g(\sigma_S^*(0, d), 0, d)$, contradicting the strict optimality of $\sigma_S^*(0, d)$ at $(0, d)$. If $u^g(\sigma_S^*(1, d)) - c^{lie}(\sigma_S^*(1, d), d) \leq u^g(\sigma_S^*(0, d)) - c^{lie}(\sigma_S^*(0, d), d)$, then (b) implies $\phi(\sigma_S^*(0, d), 1, d) \geq \phi(\sigma_S^*(1, d), 1, d)$, contradicting the strict optimality of $\sigma_S^*(1, d)$ at $(1, d)$.

(ii): We begin by showing that $\sigma_S^{*1}(1, d) \geq \sigma_S^{*2}(1, d)$. First, to simplify notation, let $\sigma^i := \sigma_S^{*i}(1, d)$, $u^g(i) := u^g(\sigma_S^{*i}(1, d))$, $k(i) := c^{lie}(\sigma_S^{*i}(1, d), d)$, and $c_j(i) := c_j^{dec}(\beta(\sigma_S^{*i}(1, d)), 1)$ (the deception cost j would incur by sending i 's message). For purposes of contradiction, suppose $\sigma^1 < \sigma^2$, which by Assumption 1 implies that 1's message is more deceptive than 2's. By strict optimality of 1's action, we have that $u^g(1) - k(1) - c_1(1) > u^g(2) - k(2) - c_1(2) \iff u^g(1) - k(1) - u^g(2) + k(2) > c_1(1) - c_1(2)$. By strict optimality of 2's action, we have that $u^g(2) - k(2) - c_2(2) > u^g(1) - k(1) - c_2(1) \iff u^g(1) - k(1) - u^g(2) + k(2) < c_2(1) - c_2(2)$. But then we have that $c_1(1) - c_1(2) < c_2(1) - c_2(2)$, which violates the assumption that 1 finds deception more costly than 2. Hence, it must be that $\sigma_S^{*1}(1, d) \geq \sigma_S^{*2}(1, d)$. A symmetric argument yields $\sigma_S^{*1}(0, d) \leq \sigma_S^{*2}(0, d)$, and therefore that $\sigma_S^{*1}(1, d) - \sigma_S^{*1}(0, d) \geq \sigma_S^{*2}(1, d) - \sigma_S^{*2}(0, d)$. $\square$

10 Online Appendix

10.1 Learning? Aggregate sender data in later rounds

Since there is no feedback, there can be no conventional learning, but we analyze later rounds to rule out that erratic initial play or exploration drives the observed effects. In Figure 14 and Table 6, we replicate the analysis of Section 4 using the last 20 rounds of play as the sender (of 40 total rounds).

Comparing Figure 14 to Figure 4, we find that the qualitative patterns are very similar in all cases. Comparing Table 6 to Table 3, we find that the estimated coin effects are slightly larger in the deception treatments (1.00, 1.29, and 1.23) and smaller in the pure lying treatment (0.20)—with the latter effect no longer significant. This implies an even larger gap between the deception and pure lying treatments, with magnitudes 5-6.5 times higher in the deception treatments relative to pure lying. Hence, if anything, the results become stronger over time.

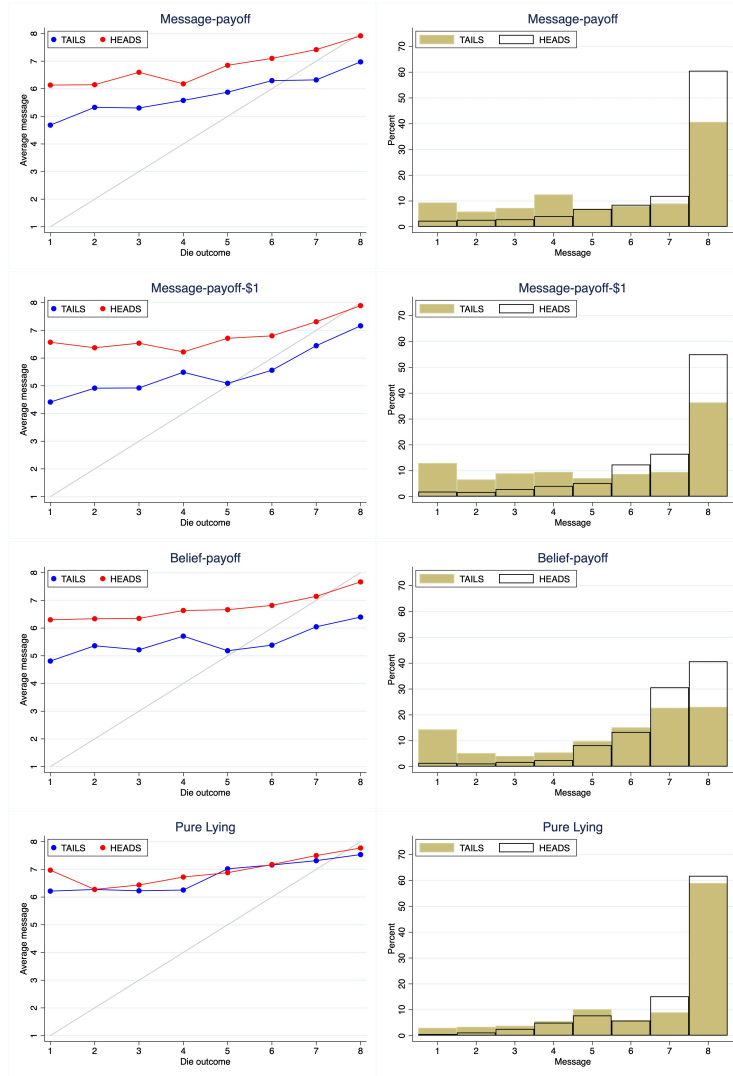


Figure 14: Average messages conditional on coin and die; distribution of messages conditional on coin (last 20 rounds).

	Coin effect				Difference in coin effects		
	MP (1)	MP-\$1 (2)	BP (3)	PL (4)	MP vs. PL (5)	MP-\$1 vs. PL (6)	BP vs. PL (7)
Heads	1.00*** (0.16)	1.29*** (0.24)	1.23*** (0.25)	0.20 (0.15)	0.20 (0.15)	0.20 (0.15)	0.20 (0.15)
Die	0.27*** (0.03)	0.26*** (0.05)	0.18*** (0.04)	0.20*** (0.05)	0.27*** (0.03)	0.26*** (0.05)	0.18*** (0.04)
Heads $\times$ treatment					0.80*** (0.22)	1.10*** (0.29)	1.04*** (0.29)
Die $\times$ PL					-0.07 (0.06)	-0.06 (0.07)	0.02 (0.06)
PL					1.28** (0.48)	1.55** (0.54)	1.12** (0.49)
Constant	4.55*** (0.27)	4.28*** (0.38)	4.71*** (0.29)	5.83*** (0.40)	4.55*** (0.27)	4.28*** (0.37)	4.71*** (0.28)
Observations	2880	1480	1640	1000	3880	2480	2640

Table 6: *Coin effects: reduced-form evidence of deception aversion (last 20 rounds).* In columns (1)-(4), for each treatment, we regress the message sent on an indicator for the coin being heads and the die outcome. Columns (1)-(4) correspond to each of the treatments. The coefficient on “Heads” is the average coin effect. In columns (5)-(7), we consider the *difference in* average coin effects between a given treatment and the pure lying treatment. The coefficient on “Heads $\times$ treatment” gives the difference in coin effects between the given treatment and pure lying. Standard errors are clustered at the subject level.

10.2 A simple model of boundedly rational beliefs

The tendency for most subjects to form monotonically increasing beliefs and for some subjects to have hump-shaped beliefs that peak at 7 (as documented in Section 3) is broadly consistent with level k models in the spirit of Wang et al. [2010]. In such a model, level 0-senders are *truthful* and level 0-receivers are *credulous*—believing that every message was sent truthfully.

The level 1-receiver, believing she faces a truthful level 0-sender, is also credulous. Hence level 0- and level 1- receivers will form beliefs that are monotonically increasing in the message due to the positive correlation between Θ and D . The level 1-sender, best responding to a level 0-receiver, will always message 8. This trivially maximizes payoffs in the message-payoff and pure lying games, and is also the payoff-maximizing action in the belief-payoff game (due to the level 0-receiver’s monotonically increasing beliefs). Now consider the level 2-receiver. Under the standard level k model, she believes she faces a level 1-sender who always message 8. However, this does not account for off-path messages that would be sent by other types. Hence, we suppose that the level 2-receiver believes she faces a large mass $(1 - \epsilon)$ of level 1-senders and a small mass ϵ of truthful level 0-senders.

Applying Bayes’ rule, this pins down the first-order beliefs b^k of the level k -receiver for $k \in \{0, 1, 2\}$. Assuming the level k -sender believes she faces a level $(k - 1)$ -receiver, this also pins down the second-order beliefs β^k of the level k -sender as $\beta^k = b^{k-1}$ for $k \in \{1, 2, 3\}$. We summarize these beliefs in Table 7.

Intuitively, when the level 2-receiver observes any message other than 8, she knows it came from a truthful level 0-sender, and so her beliefs are strictly increasing in messages 1 through 7 due to the positive correlation between Θ and D . Instead, when the level 2-receiver observes message 8, she knows it may have come from a level 1-sender who always messages 8, independent of the coin. A message of 8 may therefore be a relatively weak signal that the coin was heads, depending on $(1 - \epsilon)$ —the fraction of level 1-senders she believes she faces. If $(1 - \epsilon)$ is sufficiently high, then her beliefs will be hump-shaped with a peak at 7. Using the expressions from Table 7, the exact condition for hump-shaped beliefs is $b^{k=2}(8; \epsilon) < b^{k=2}(7; \epsilon) \iff (1 - \epsilon) > \frac{1}{21}$. Similarly, because $\beta^k = b^{k-1}$, the level 3-sender’s second-order beliefs will be hump-shaped if and only if $(1 - \epsilon) > \frac{1}{21}$.

k	Sender's second-order beliefs $\beta^k(m; \epsilon)$	Receiver's first-order beliefs $b^k(m; \epsilon)$
0	—	$0.32 + 0.04m$
1	$0.32 + 0.04m$	$0.32 + 0.04m$
2	$0.32 + 0.04m$	$\frac{\epsilon(0.08+0.01m)+(1-\epsilon)\mathbf{1}\{m=8\}}{\epsilon 0.25+2(1-\epsilon)\mathbf{1}\{m=8\}}$
3	$\frac{\epsilon(0.08+0.01m)+(1-\epsilon)\mathbf{1}\{m=8\}}{\epsilon 0.25+2(1-\epsilon)\mathbf{1}\{m=8\}}$	...

Table 7: *Level k beliefs.*

We take the fact that most subjects have monotonic beliefs and that the second-largest group have hump-shaped beliefs as strong evidence for the type of credulity embodied in the level k model. Of course, no subjects conform to any of the level k -types perfectly. For instance, an exactly level 0- or level 1-receiver would form beliefs that correspond to the *truthful/Bayesian* benchmark (black line) in Figures 2 and 3. The only systematic deviation we observe is the tendency to *over-update*: relative to the truthful/Bayesian benchmark, we see that subjects tend to over-update in the “tails direction” following messages 1-4 and in the “heads direction” following messages 5-8. This may either be an overreaction to signals, or it may simply be that subjects believe high messages are sent more often following heads and less following tails than implied by truth-telling. Both are plausible, so we do not take this as evidence against the credulity mechanism.

We conclude this section with three remarks.

Remark 7. The level 1-sender’s second-order beliefs coincide with the truthful/Bayesian benchmark. Hence, the sender’s payoffs in the message-payoff game, $u^M(m) = 0.32 + 0.04m$, exactly coincide with those the level 1-sender believes she faces in the belief-payoff game: $u^B(\beta^{k=1}(m)) = \beta^{k=1}(m) = 0.32 + 0.04m$. Hence, if level k is the true model generating beliefs, then the message-payoff and belief-payoff games are similar for subjects with low levels of sophistication.

Remark 8. In the message-payoff game, the receiver’s beliefs do not matter for the sender’s best response. As such, all senders of level $k \geq 1$ will send message 8 with probability 1. In the belief-payoff game, however, because the sender is incentivized to maximize the receiver’s beliefs, the level 3-sender will message 7 when her second-order beliefs are hump-shaped. Indeed, senders of even higher level, if they believe they face

a distribution of lower level receivers, may message lower numbers yet. Hence, the level k model captures the intuitive idea that a message of 8 is a “likely story” or “too good to be true,” and so lower messages may be used by more sophisticated subjects. This leads to the prediction that senders will send lower messages in the belief-payoff game, which is exactly what we observe (see, for example, Figure 4).

Remark 9. The level k model is closely related to the family of beliefs β_α used in the structural model of Section 7. We think of $\beta_{\alpha=0}$ as the second-order beliefs of a level 1- or level 2-sender who believes the receiver is perfectly credulous (i.e. level 0 or 1), except the assumption of perfect Bayesianism is dropped in favor of belief updating more consistent with the data. Similarly, we think of $\beta_{\alpha=1}$ as the second-order beliefs of a level 3-sender who believes the receiver is level 2. Intermediate values of α interpolate between these two extreme second-order belief functions, and so higher α indicates a higher degree of strategic sophistication.

10.3 Unexplained patterns in sender strategies

A small number of subjects exhibit die effects that cannot be captured qualitatively by the structural model, suggesting fixed or other non-linear lying costs. In Figure 15, we show a few examples. Subject MP10 tells the truth for die outcomes 5-8, and otherwise messages 8. This suggests a *fixed cost* associated with lying, such that it is optimal to tell the truth if it leads to good outcomes and tell a *maximal lie* otherwise. BP10 tells the truth for die outcomes 5-8, and messages $d + 4$ for die outcomes $d \in \{1, \dots, 4\}$. This is consistent with two tiers of fixed costs: one associated with any degree of lying, and another that applies only if the distance between the true die outcome and message exceeds 4. BP11 messages 6 for die outcomes 1-6, and tells the truth otherwise. This can be explained by incorporating image concerns—i.e. a cost of being perceived as a liar in the spirit of Gneezy et al. [2018] and Abeler et al. [2019] (see also footnote 18). That is, if the subject thinks that messaging 7 and higher signals that she is lying with high probability, she may avoid such messages unless they are true. Alternatively, it may also be that the subject has an “aspiration level” and wishes to guarantee at least the payoff associated with 6.

There may also be some unexplained deception aversion. If the structural model per-

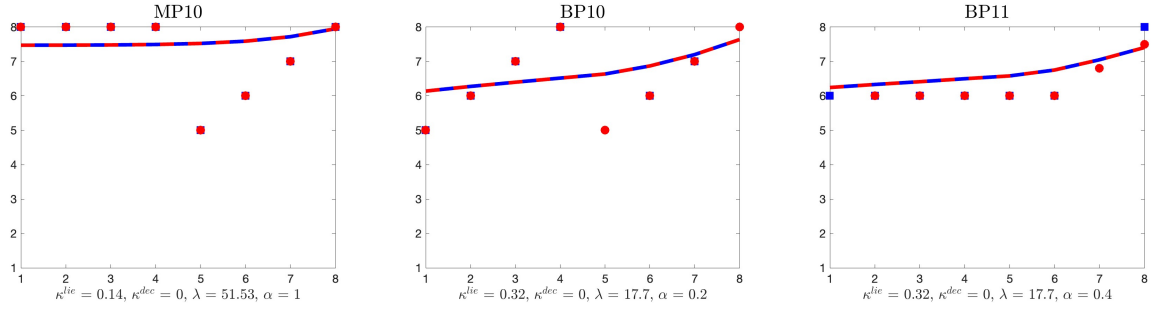


Figure 15: *Unexplained lying aversion—individual subjects.*

fectly captured subjects' utility functions and second-order beliefs, then we would expect (up to some randomness) no coin effects among subjects classified as non-deception-averse—those for which the best model involves $\kappa^{dec} = 0$. However, among such subjects, we identify three (all from belief-payoff) with C_i exceeding 1.20 (the next highest being 1.01). We give plots for these subjects in Figure 16. These subjects have clear positive coin effects, but the model provides a poor fit and does not identify them as deception-averse. In our view, these subjects most likely *are* deception-averse, but the model is not flexible enough to capture the exact form in which it manifests. These examples suggest that the structural estimates may understate the degree of deception aversion, further highlighting the value of the reduced-form analysis.⁴⁷

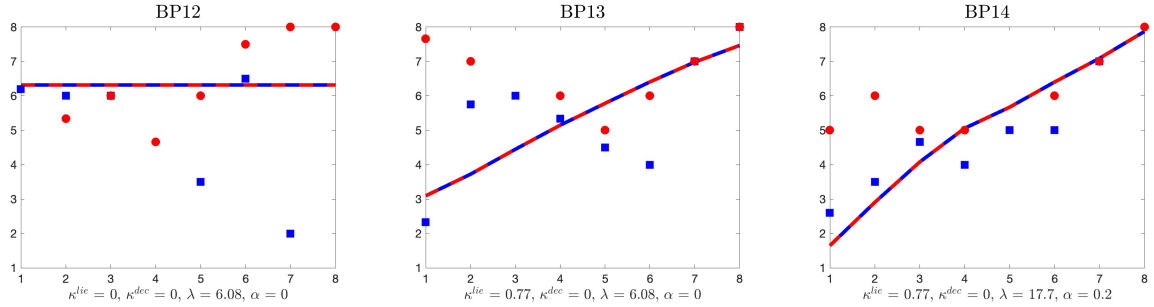


Figure 16: *Unexplained deception aversion—individual subjects.*

⁴⁷Note also that, because the structural model assumes monotonically increasing or hump-shaped second-order beliefs that peak at 7, it classifies the small number of subjects with negative coin effects as non-deception-averse. In principle, however, such subjects may be deception-loving or deception-averse with *decreasing* second-order beliefs. Of course, they may also be confused or careless.

We observe some other patterns in a handful of individual subjects' data that we believe result from deliberate and intuitive strategies, but are still not well captured by the structural model. Overall, our sense is that more subjects in the belief-payoff treatment display such patterns, perhaps because the game's higher complexity leads to greater use of simple heuristics. In Figure 17, we give the data for three such subjects, all from the belief-payoff treatment.

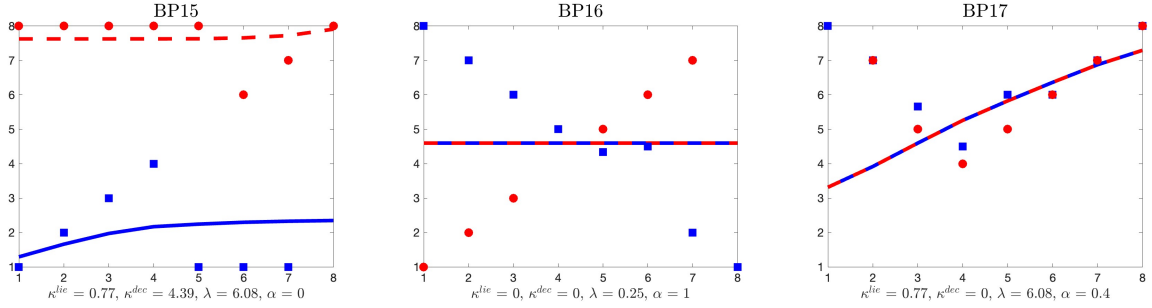


Figure 17: *Other unexplained patterns.*

Referring to Figure 17, subject BP15 is classified as deception-averse, but exhibits a pattern that cannot be explained by the structural model. Following heads, this subject tells the truth for die outcomes 6-8, but messages 8 otherwise; following tails, the subject tells the truth for die outcomes 1-4, but messages 1 otherwise. Her overall behavior suggests a heuristic of “tell the truth if deception is below some threshold; minimize deception otherwise.” Because this subject exhibits both coin and die effects, we know that both lying and deception costs are needed to rationalize her behavior. This pattern can be explained by a combination of lying costs and a *fixed cost* from deceiving above some threshold. BP16 tells the truth following heads and tells the opposite of the truth following tails, i.e. $m = 9 - d$. Recalling that incentives are to convince the receiver that the coin was heads, this subject is “telling the truth in the good state; saying the opposite in the bad state.” Hence, while misguided in this context, this may be an application of an intuitive heuristic, as opposed to confusion about the game per se. BP17 tells the opposite of the truth for die outcomes 1-4 and tells the truth for die outcomes 5-8. Hence, she applies a similar heuristic: “tell the truth if (she believes) it leads to good outcomes; say the opposite otherwise.”

10.4 Robustness to alternative structural models

We consider four alternative structural models and conduct the same exercise from Section 7 to type individual subjects.

The first three models are the same as in Section 7 (i.e. with linear lying and deception costs), except we use alternative specifications for the second-order belief function. Model (A1) uses the average elicited beliefs of *all* subjects.⁴⁸ Model (A2) uses the “fictitious” second-order beliefs $\beta(m) = 0.32 + 0.04m$ that are implied by the truthful/Bayesian benchmark. Model (A3) uses, for each individual subject, her elicited beliefs. Noting that we elicit first-order beliefs for some subjects and second-order beliefs for others, for some subjects, we use first-order beliefs to proxy for second-order.

Model	Game	Averse to:			
		Lying and deception ($\kappa^{lie} > 0, \kappa^{dec} > 0$)	Deception only ($\kappa^{lie} = 0, \kappa^{dec} > 0$)	Lying only ($\kappa^{lie} > 0, \kappa^{dec} = 0$)	Neither ($\kappa^{lie} = \kappa^{dec} = 0$)
(A1)	Message-payoff	15% (21)	13% (18)	35% (51)	38% (54)
	Message-payoff-\$1	22% (16)	11% (8)	38% (28)	30% (22)
	Belief-payoff	15% (12)	12% (10)	48% (39)	26% (21)
	Pure lying	4% (2)	4% (2)	34% (17)	58% (29)
(A2)	Message-payoff	16% (23)	13% (18)	35% (50)	37% (53)
	Message-payoff-\$1	23% (17)	8% (6)	38% (28)	31% (23)
	Belief-payoff	13% (11)	13% (11)	40% (33)	33% (27)
	Pure lying	2% (1)	6% (3)	34% (17)	58% (29)
(A3)	Message-payoff	13% (19)	12% (17)	38% (54)	38% (54)
	Message-payoff-\$1	22% (16)	11% (8)	36% (27)	31% (23)
	Belief-payoff	7% (6)	13% (11)	35% (29)	44% (36)
	Pure lying	0% (0)	6% (3)	36% (18)	58% (29)
(A4)	Message-payoff	9% (13)	23% (33)	31% (44)	38% (54)
	Message-payoff-\$1	15% (11)	20% (15)	28% (21)	36% (27)
	Belief-payoff	10% (8)	21% (17)	26% (21)	44% (36)
	Pure lying	6% (3)	10% (5)	26% (13)	58% (29)

Table 8: *Summary of structural estimates—Robustness.*

Model (A4) is the same as the model from Section 7, except the lying and deception costs are assumed to be quadratic: $c^{lie}(m, d) = \kappa^{lie}((m - d)/7)^2$ and $c^{dec}(\theta, \beta(m)) =$

⁴⁸That is $\beta(1, 2, \dots, 8) = (0.229, 0.266, 0.318, 0.401, 0.519, 0.609, 0.689, 0.700)$, which is strictly monotonic in messages.

$\kappa^{dec}(\beta(m) - \theta)^2$, respectively.

Table 8, which is directly comparable to Table 5, summarizes the results. The alternative models imply similar distributions of types. In particular, the estimated fraction of deception-averse individuals in the deception treatments varies only between 26 to 32% across all five models, and so we conclude the results are robust.

10.5 Comparing elicited beliefs across treatments

	Belief
	(1)
MP	-1.707 (1.832)
MP-2nd	-3.414** (1.608)
MP-\$1	-0.414 (1.694)
BP	1.024 (1.735)
Constant	24.255*** (1.564)
Observations	3100
p-value: MP = MP-2nd = MP-\$1 = BP	0.117

Table 9: *Comparing elicited beliefs across treatments.* We regress elicited beliefs on indicators for message-payoff, message-payoff-2nd-order, message-payoff-\$1, and belief-payoff (with pure lying as the omitted category) and dummy variables for each message (omitted from the table). Standard errors are clustered at the subject level. We perform an F -test of the hypothesis that all 4 coefficients are 0, i.e. that average beliefs are the same across all five treatments. We cannot reject, meaning that average beliefs are similar across treatments.

10.6 Experimental instructions

We present the experimental instructions for the message-payoff treatment. Instructions for the other treatments were designed to be as similar as possible, and are available from the authors on request.

Experimental Instructions

The experiment will take approximately **50 minutes**, and you will receive **\$10** just for completing the experiment. In addition, you may receive additional payments based on your choices and the choices of other participants.

Please turn off your cellphones now. The session will be fairly short, so please try to be attentive throughout. Please feel free to ask questions at any point during the instruction period or during the experiment itself.

The experiment will have **one main section**, followed by a short **questionnaire**.

The main section, which we will describe now, has **41 rounds** in total.

Rounds 1-40

In each of the first 40 rounds, you will privately observe the outcome of a **coin flip**, which will be either HEADS or TAILS. Both are equally likely (50-50 chance of HEADS or TAILS).

After observing the outcome of the coin, you will also privately observe the outcome of a **roll of an 8-sided die**. The die is *weighted*:

- If the coin was HEADS, you are *more likely to roll **higher** numbers*.
- If the coin was TAILS, you are *more likely to roll **lower** numbers*.
- If the coin was HEADS, the exact probabilities of die outcomes are given in the following table:

Die outcome:	1	2	3	4	5	6	7	8
Probability:	9%	10%	11%	12%	13%	14%	15%	16%

- If the coin was TAILS, the exact probabilities of die outcomes are given in the following table:

Die outcome:	1	2	3	4	5	6	7	8
Probability:	16%	15%	14%	13%	12%	11%	10%	9%

In each of the first 40 rounds, you will flip a new coin and roll a new die. The outcomes in each round will be completely independent of previous rounds' outcomes AND completely independent of other participants' outcomes. For the entire experiment, no participant (other than you) will ever observe the outcomes of your coin flips or your die rolls.

In each round, after seeing your coin flip and die roll, you will **send a message**. Your message will say:

<i>"The outcome of my die roll is ____"</i>

where you must fill the blank with any number 1, 2, 3, 4, 5, 6, 7, or 8.

No one will see any of your messages until *after* all 40 rounds. In round 41, **one round** from rounds 1-40 will be randomly selected (all equally likely), and the message you sent in that round will be shown to another randomly selected participant.

We will describe the exact details later, but the **payment** you receive for sending messages will depend *only* on your randomly selected message, and you will tend to *earn more money* by reporting higher die outcomes.

NOTE: You may report any number you wish, independent of the actual outcome of the die roll.

Round 41

In round 41, you will observe the message sent by a randomly selected participant (other than you) in a randomly selected previous round (1-40). After observing the message (1, 2, 3, 4, 5, 6, 7, or 8), you will **report your guess** for the probability the participant's coin was HEADS in the randomly selected round. That is, you will be asked:

A participant (other than you) and a round are randomly selected.

In this round, the participant sent the message:

“The outcome of my die roll is ____”

What is your guess for the probability this participant’s coin was HEADS in this round?

You will then enter your numerical guess between 0 and 100 via a slider.

NOTE: If you had no information, your best guess would be 50% (since the coin gives an equal chance of HEADS and TAILS). However, the message sent by the participant may give you additional information, allowing you to make a better guess.

Payments

Payment for sending messages (Rounds 1-40). Recall that, in round 41, one of your messages will be randomly selected and shown to another participant (other than you).

Depending on your message that is randomly selected, *you will receive* \$10 with some probability. This probability depends *only* on your message. The exact probabilities are given in the following table:

Randomly selected message:	1	2	3	4	5	6	7	8
Your probability of receiving \$10:	36%	40%	44%	48%	52%	56%	60%	64%

For example,

- If your randomly selected message is 2, you will receive \$10 with probability 40%.
- If your randomly selected message is 6, you will receive \$10 with probability 56%.

Hence, you will tend to *earn more money* by reporting higher die outcomes.

Payment for making a guess (Round 41). Recall that, after observing the message sent by a randomly selected participant (other than you), you will report your guess for the probability that the participant's coin was HEADS.

It is in your best interest to simply tell us your best guess. To ensure this, we use the following **payment procedure**:

Your guess is a number X between 0 and 100 that gives the probability that you believe the participant's coin was HEADS.

(1) If the participant's coin was HEADS, then you receive \$10 with probability $\left[100 - \frac{(100-X)^2}{100}\right] \%$

(2) If the participant's coin was TAILS, then you receive \$10 with probability $\left[100 - \frac{X^2}{100}\right] \%$

This procedure may seem complicated, but don't worry! You do not need to understand the details. You just need to know that, if you believe that the probability of HEADS is $X\%$, you will maximize the probability of earning \$10 by reporting exactly $X\%$.

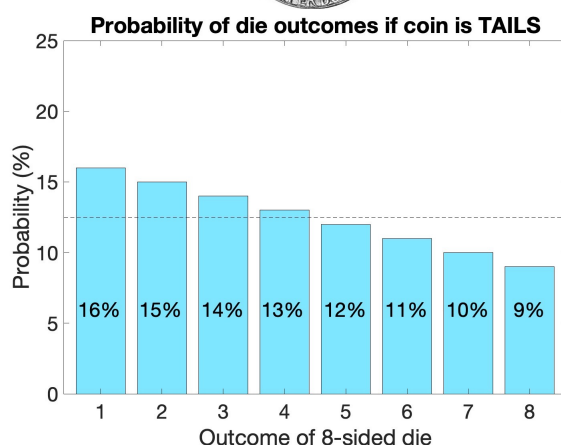
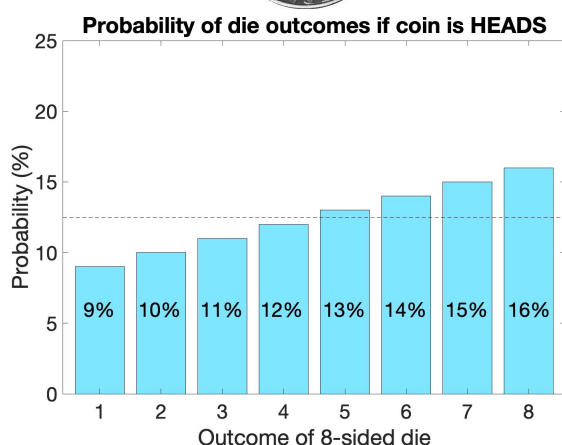
Hence, *whatever your best guess is, it is in your best interest to simply tell us.*

[Subjects are also provided with the following 1-page summary sheet]

Summary

Rounds 1-40. In each round, ...

- You will observe the outcomes of a **coin** and an **8-sided die**
 - If the coin is HEADS, you are more likely to roll higher numbers
 - If the coin is TAILS, you are more likely to roll lower numbers



- After observing the outcome of the coin and die, you will send a message reporting the outcome of the die, but you may send whatever message you like. *No participant (other than you) will ever observe the outcomes of your coin flips or your die rolls.*
- One of your 40 messages will be randomly selected and shown to another participant in round 41.

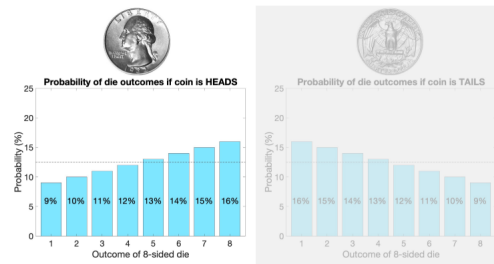
Round 41. You will observe the message sent by a randomly selected participant (other than you) and make a guess for the probability that the participant's coin was HEADS in the round when they sent that message.

Payment.

- Sending a message. By reporting higher die outcomes, you will increase the probability of receiving \$10. The exact probabilities are given on Page 3 of the instructions.
- Making a guess. You will maximize your probability of earning \$10 by simply reporting your exact best guess.

10.7 Experimental screenshots

Round 12 of 41



The outcome of the coin flip is **HEADS**.

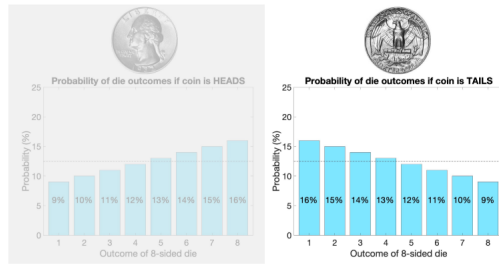
The outcome of the die roll is **7**.

What message would you like to send?

"The outcome of my die roll is ..."

Figure 18: *Rounds 1-40 (heads).* In a round where the coin outcome is heads, the tails-conditional distribution of die outcomes is grayed out, but still visible, to highlight the relevant distribution while still displaying the ex-ante joint distribution of the coin and die. Subjects are asked to report a message.

Round 21 of 41



The outcome of the coin flip is **TAILS**.

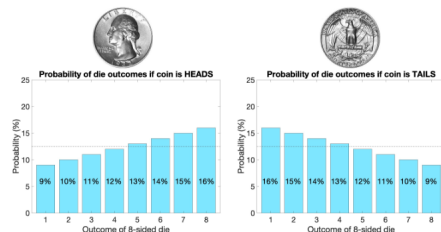
The outcome of the die roll is **2**.

What message would you like to send?

"The outcome of my die roll is ..."

Figure 19: *Rounds 1-40 (tails).* In a round where the coin outcome is tails, the heads-conditional distribution of die outcomes is grayed out, but still visible, to highlight the relevant distribution while still displaying the ex-ante joint distribution of the coin and die. Subjects are asked to report a message.

Round 41 of 41



A participant (other than you) and a round are randomly selected.
In this round, the participant sent the message:

"The outcome of my die roll is **7**."

What is your guess for the probability this participant's coin was HEADS in this round?

Click the blue bar to reveal the slider.

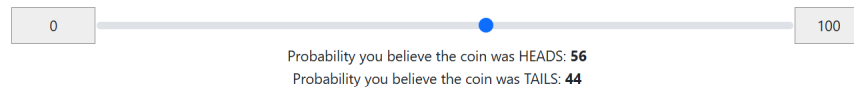
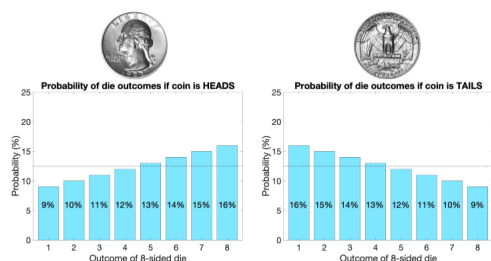


Figure 20: *Round 41 (deception treatments).* The message of a randomly selected other subject is shown, and subjects enter their beliefs about the coin via a slider. The slider is "activated" by clicking on it and has no default position.

Round 41 of 41



A participant (other than you) and a round are randomly selected.
In this round, the participant sent the message:

"The outcome of my die roll is 5."

Please click "Next" to acknowledge you have seen the message and continue the experiment.

Next

Figure 21: *Round 41 (pure lying).* The message of a randomly selected other subject is shown, and subjects simply have to click a button to acknowledge having seen the message.

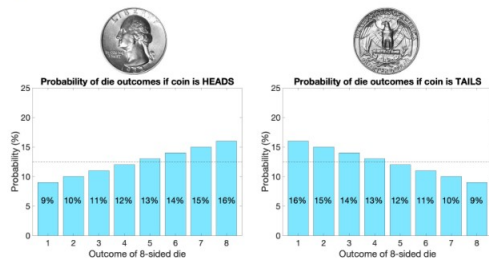
Questions 1-8

Imagine it is again round 41. We wish to know what *would have been* your guess for the probability the other participant's coin was HEADS if you observed different messages.

Please answer the following 8 questions. This does not count toward your payment, but *please answer carefully*.

Next

Question 1 of 25

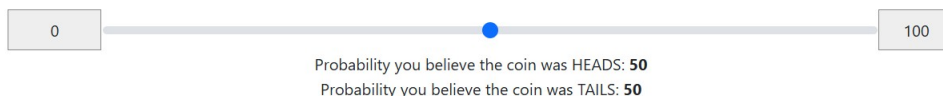


A participant (other than you) and a round are randomly selected.
In this round, the participant sent the message:

"The outcome of my die roll is 5."

What is your guess for the probability this participant's coin was HEADS in this round?

Click the blue bar to reveal the slider.



Next

Figure 22: *First-order belief elicitation.* The eight first-order belief elicitation questions look identical to that in round 41 of the deception treatments, except that there is one for each of the eight messages, appearing on separate screens and in random order.

Questions 1-8

Recall that, in round 41, each participant observed one message and gave their best guess for the probability that the other participant's coin was HEADS in the round when that message was sent.

For each of the 8 messages, we will ask you for what you think is the *average guess* (between 0 and 100) that was made among all participants (other than you) who observed that message.

It is in your best interest to simply tell us exactly what you think is the *average guess of other participants*. To ensure this, we use a **payment procedure** that is very similar to the one you saw before. By telling us exactly what you think, you maximize your chance of earning \$1. You may see the details by clicking the box below.

See payment details

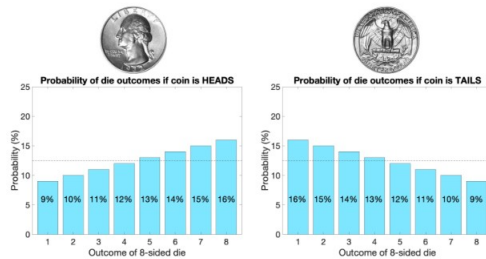
For each of the 8 messages, you will tell us what you think is the *average guess* of all participants (other than you) who observed that message in round 41.

We will then randomly choose (i) one of the 8 messages and (ii) the guess of another participant (who observed that specific message). If you tell us the average guess is X and the randomly drawn participant's guess is Y (both numbers between 0 and 100), you will receive \$1 with probability $(100 - (X - Y)^2 / 100) \%$

Close

Next

Question 1 of 25

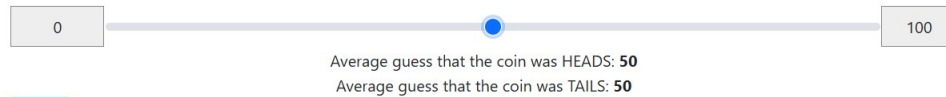


Consider all participants (other than you) who received the message:

"The outcome of my die roll is 2"

What do you think is the *average guess* made by these participants about the probability that the coin was HEADS in the round when this message was sent?

Click the blue bar to reveal the slider.



Next

Figure 23: *Second-order belief elicitation.* The eight second-order belief elicitation questions look as close as possible to that in round 41 of the deception treatments, except that there is one for each of the eight possible messages, appearing on separate screens and in random order.